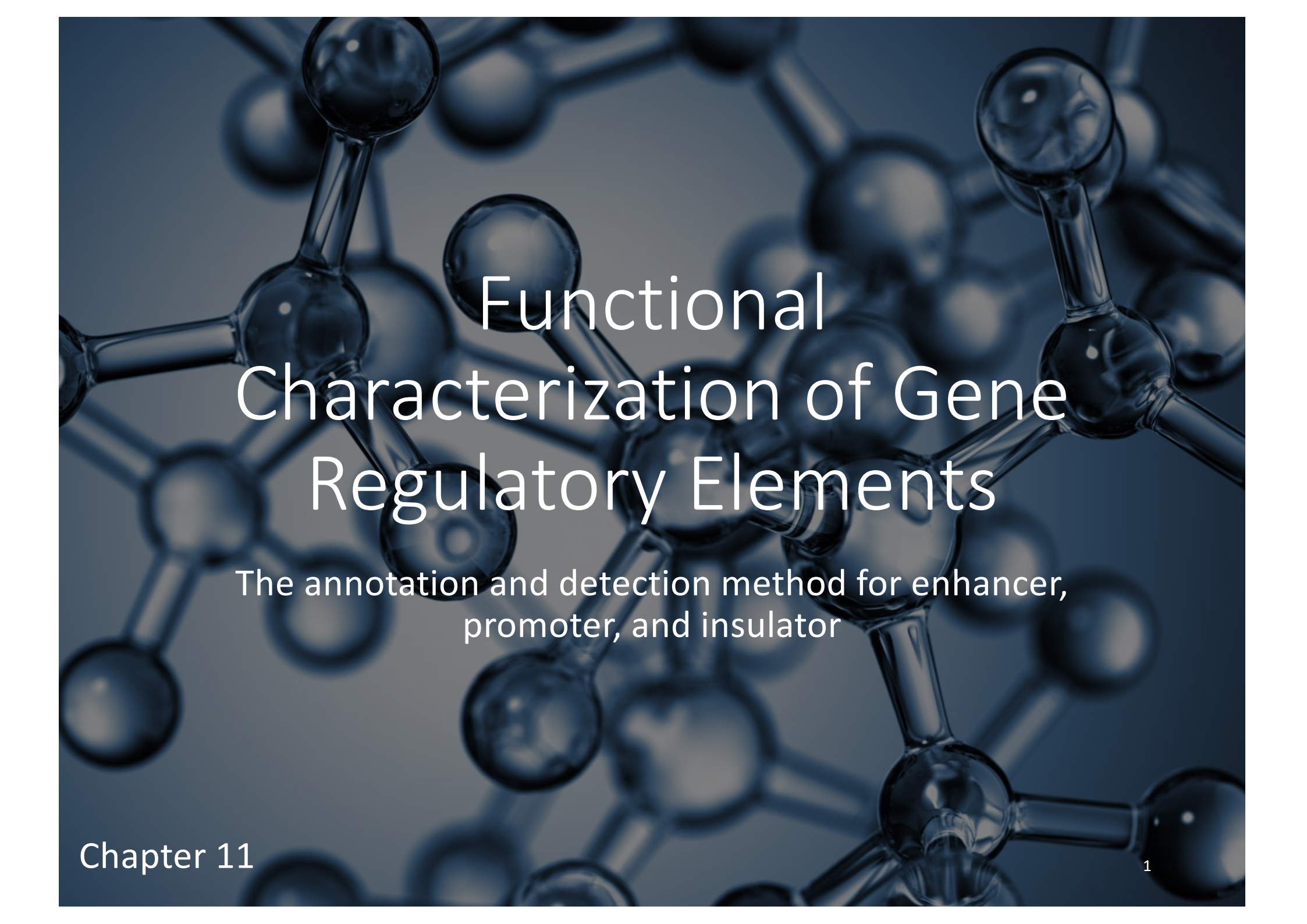


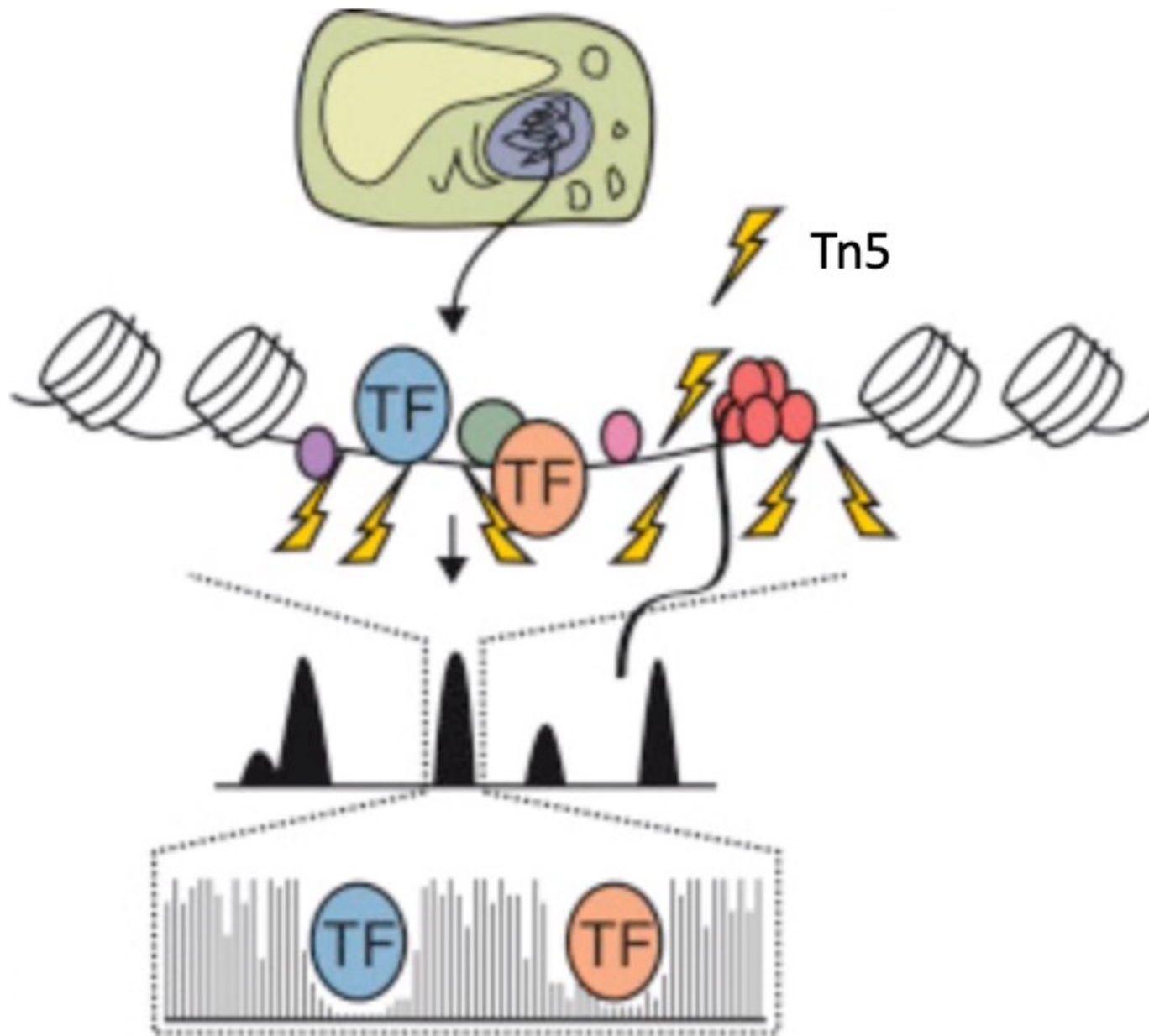


Functional Characterization of Gene Regulatory Elements

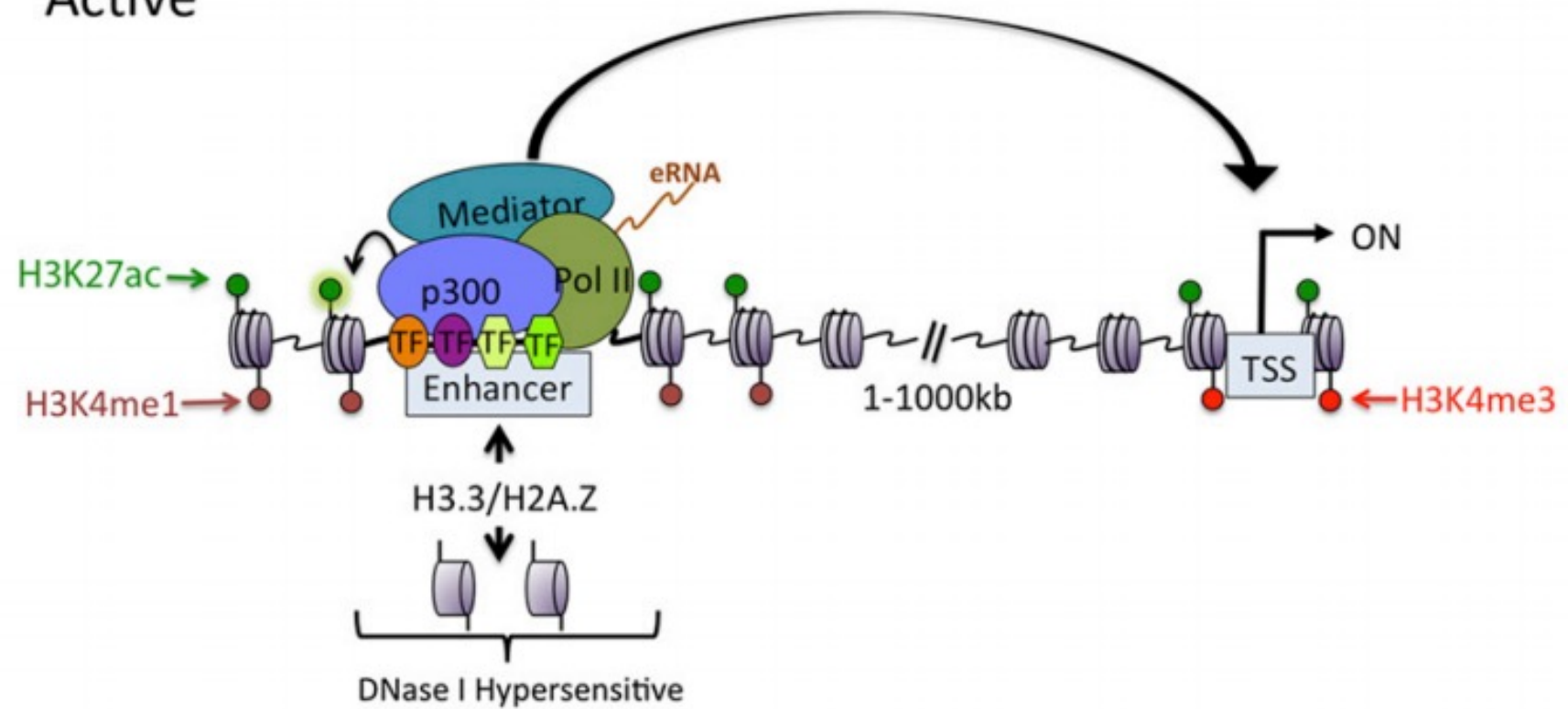
The annotation and detection method for enhancer,
promoter, and insulator

Main content

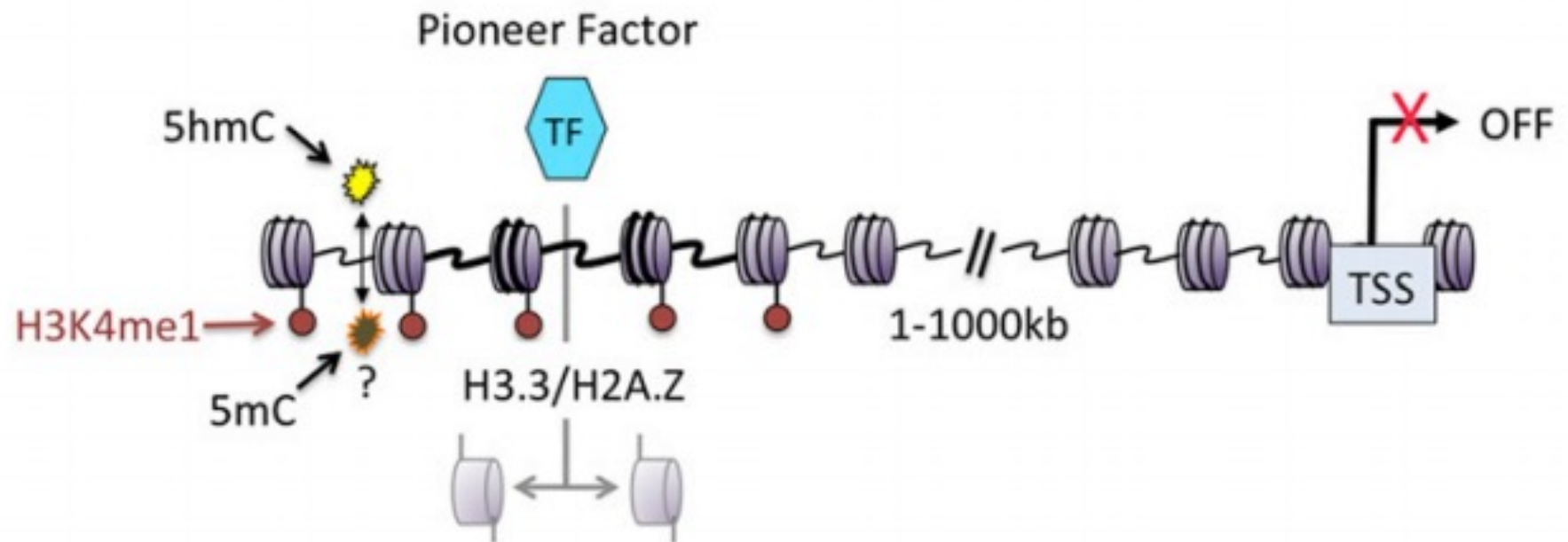
- ChIP-Seq
- CUT&RUN
- STARR-Seq



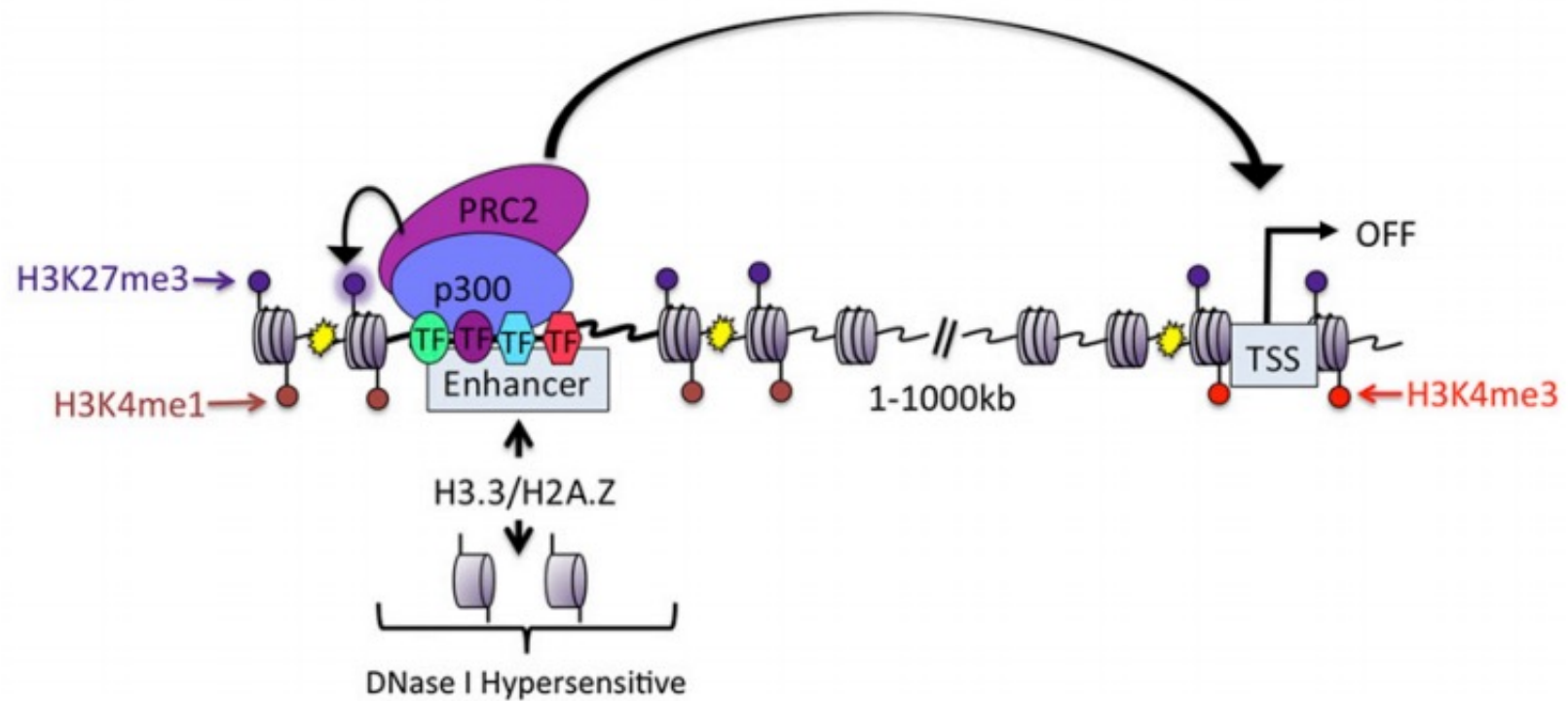
Active



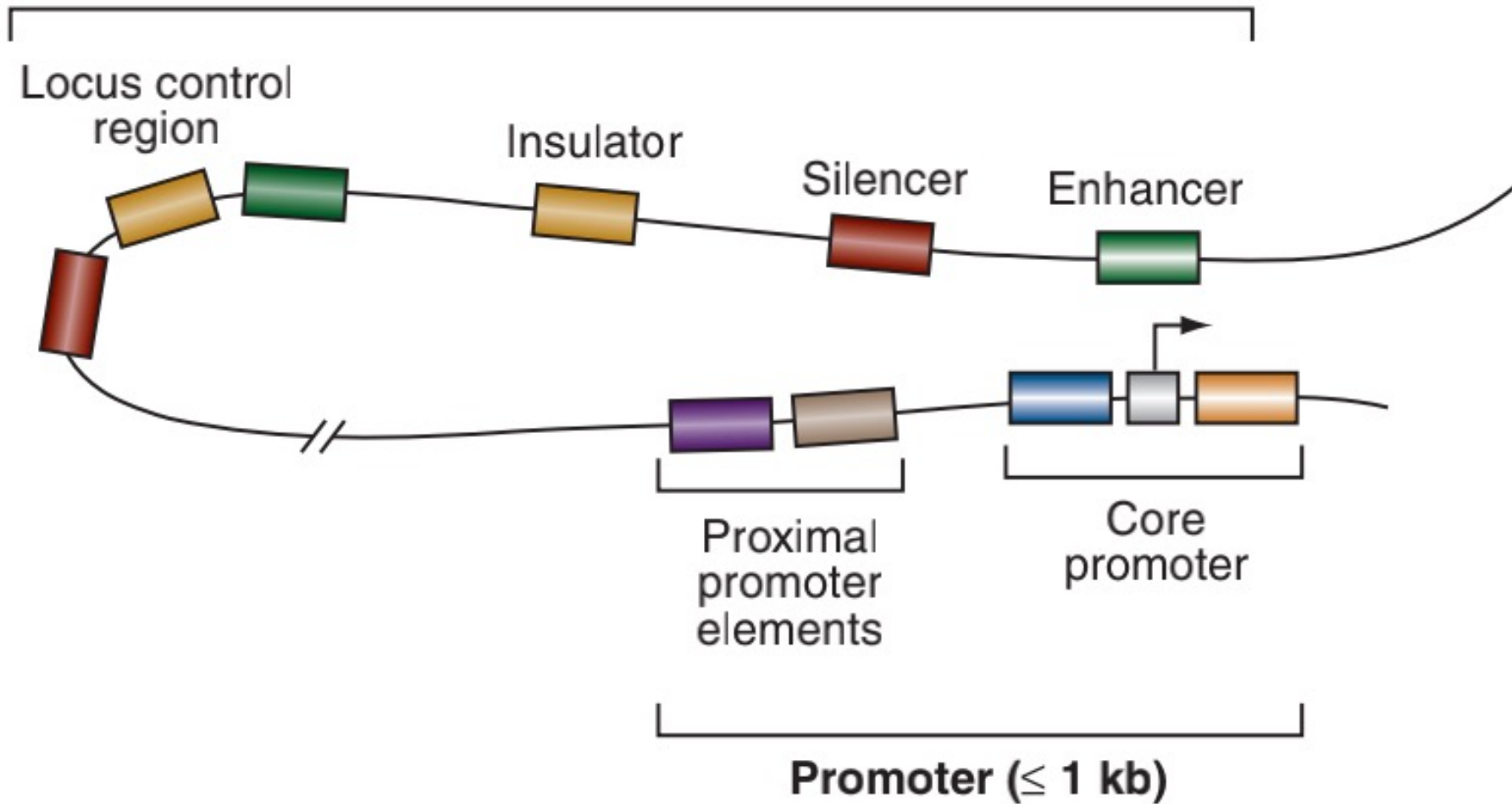
Primed



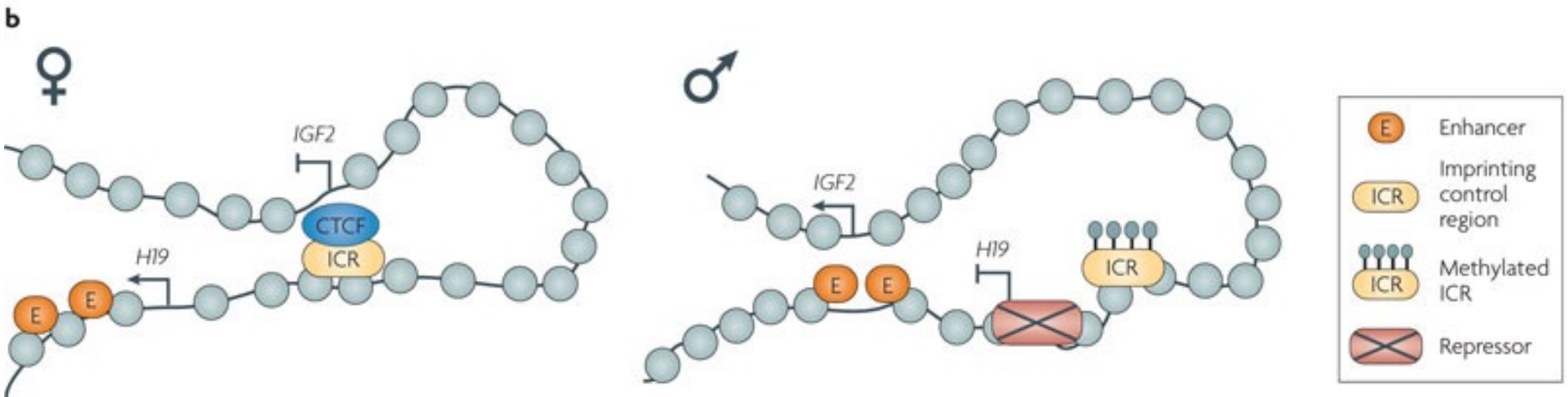
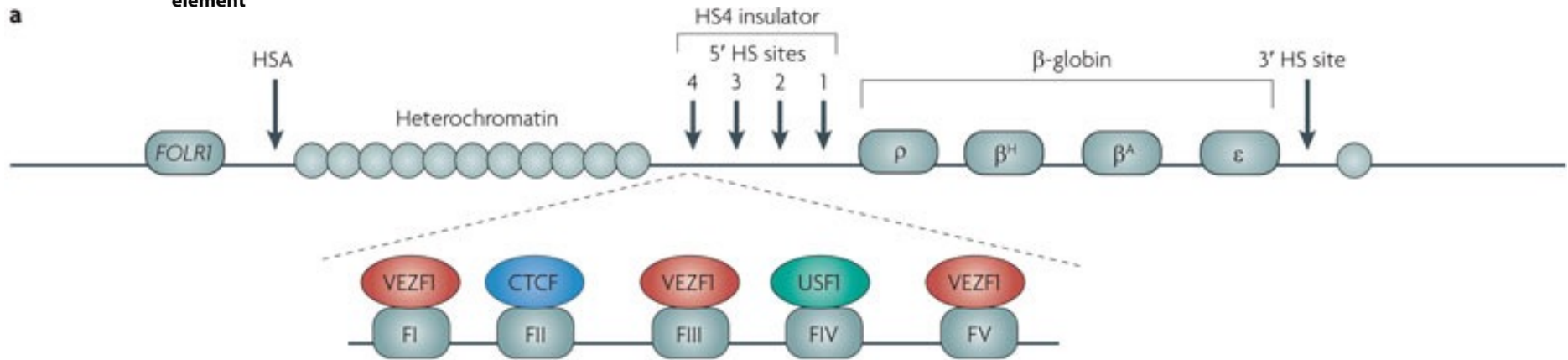
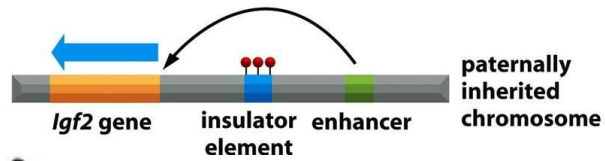
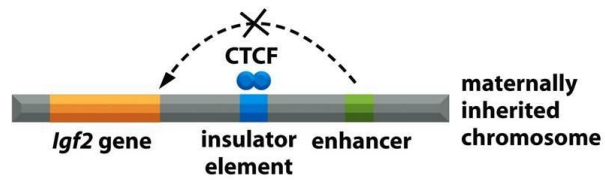
Poised (mouse and human ESC)



Distal regulatory elements

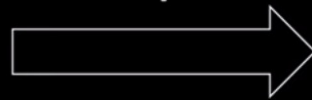


Transcriptional Regulatory
Elements in the Human
Genome



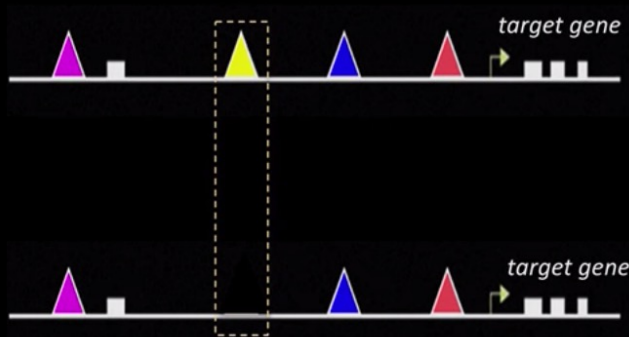
cis-regulatory
variation

development



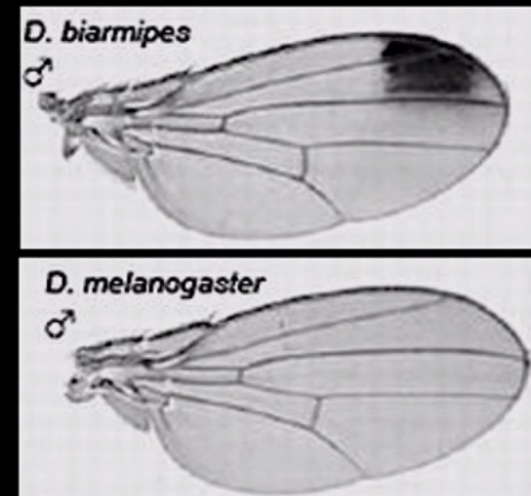
phenotype
variation

enhancers



Sean Carroll, David Kingsley, Mike Levine & others

wing patterning in Drosophila

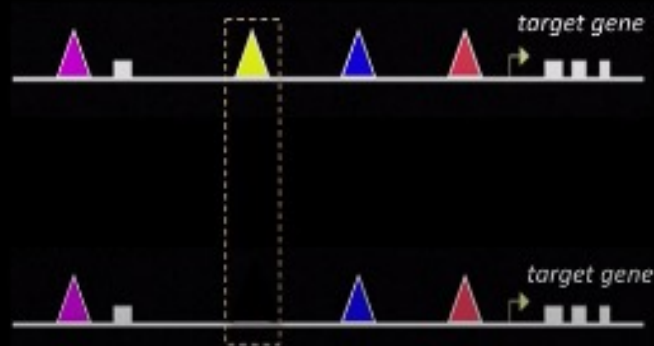


cis-regulatory
variation



phenotype
variation

enhancers



human iPSC

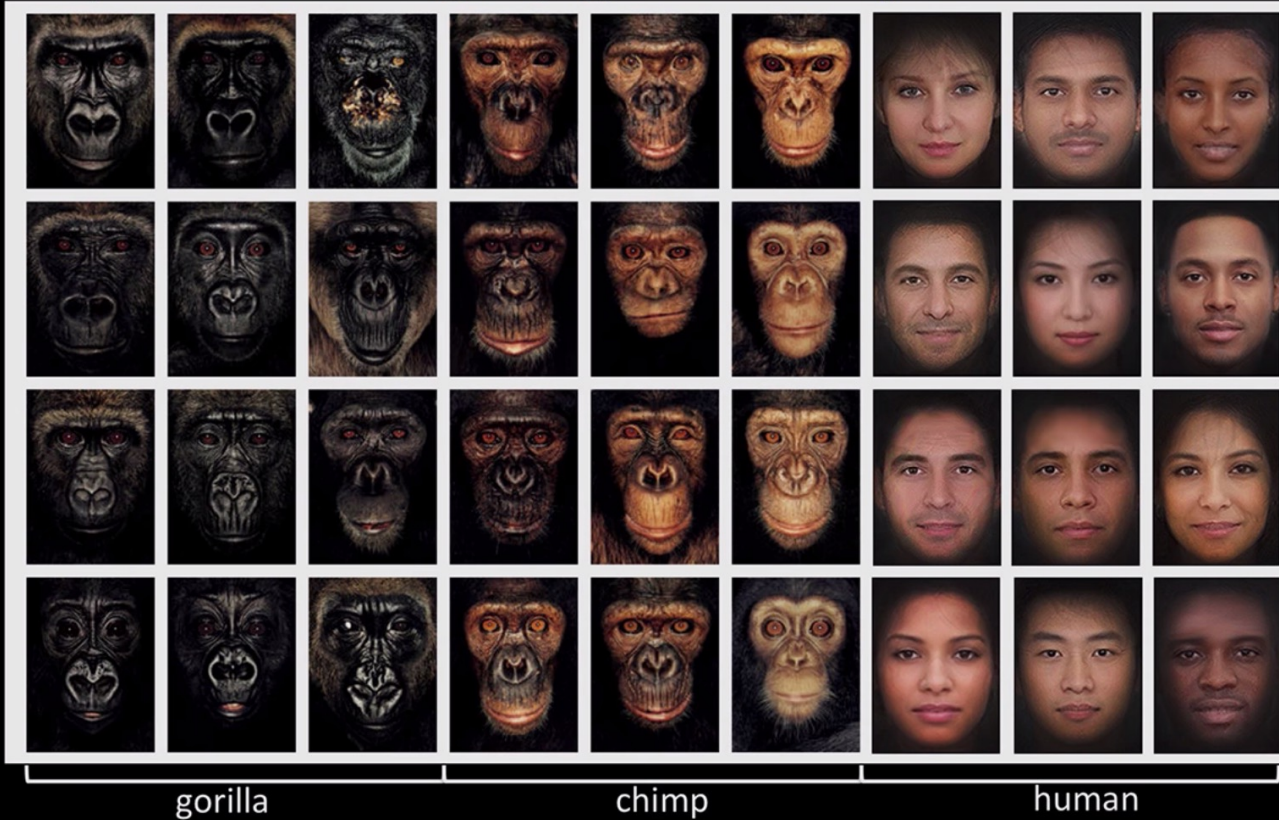


chimp iPSC

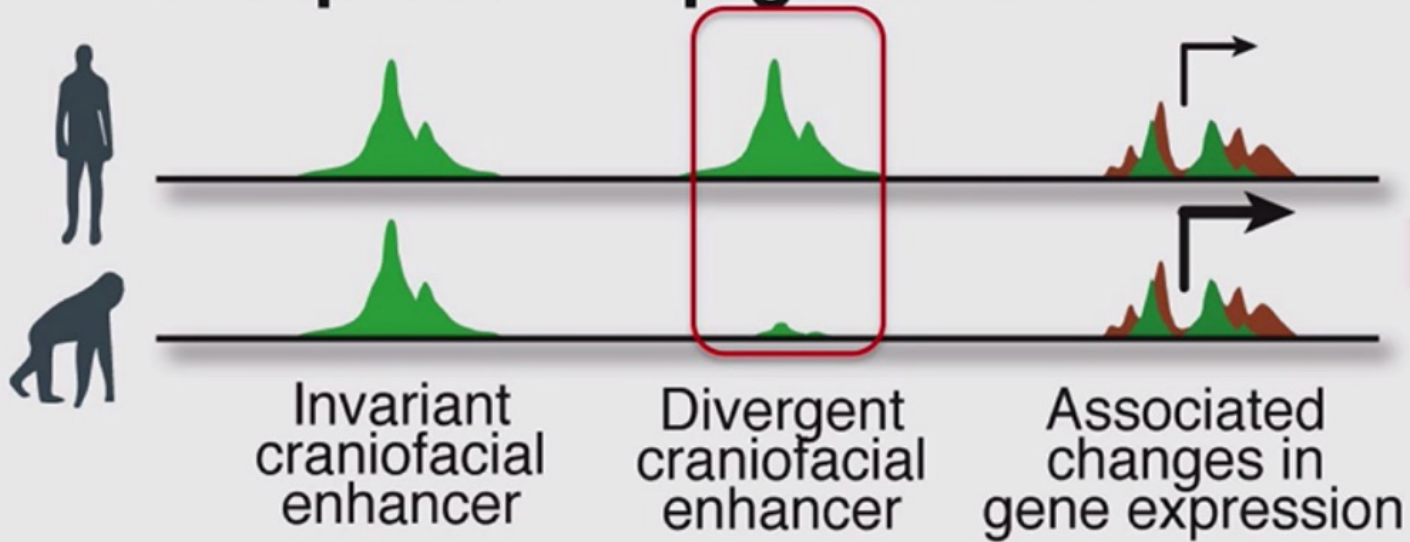


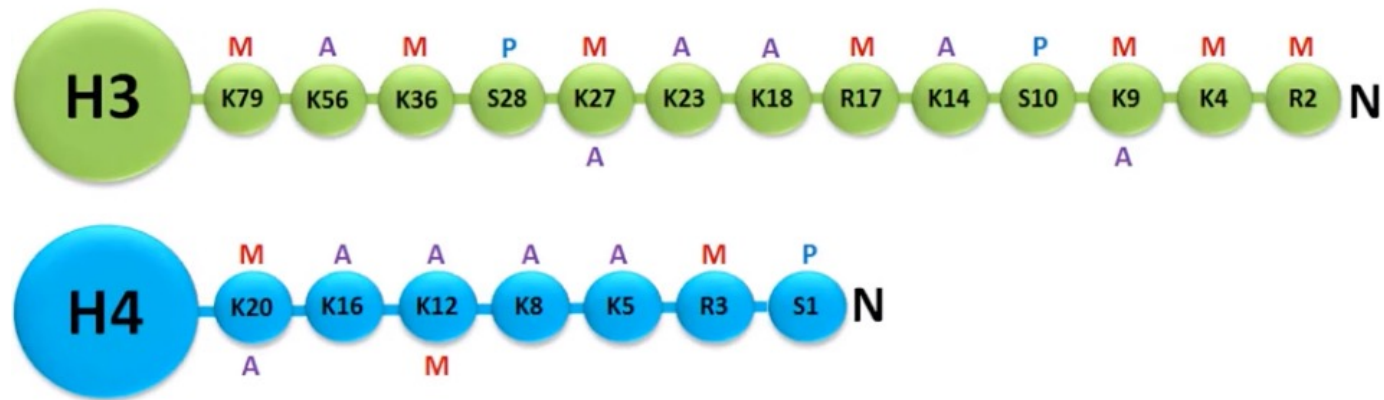
Prescott et al, *Cell*, 2015

Craniofacial variation within and between species is largely genetically encoded

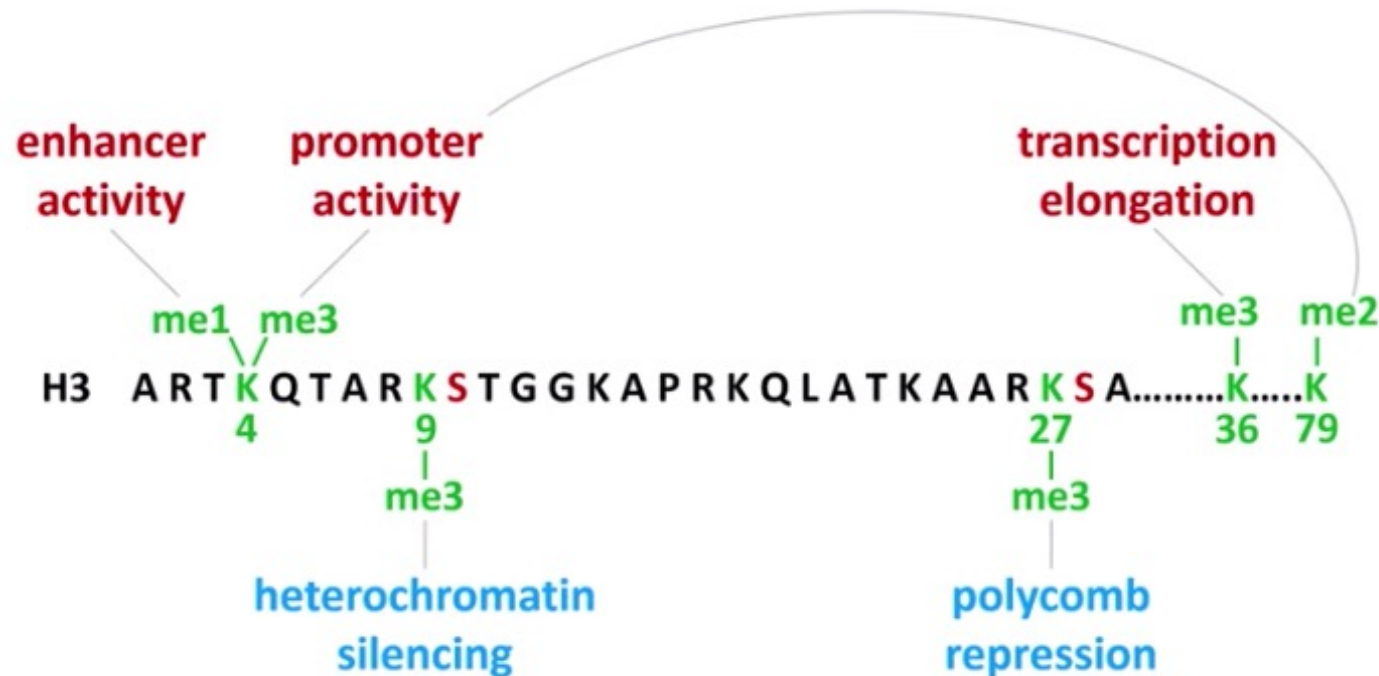


Comparative Epigenomics

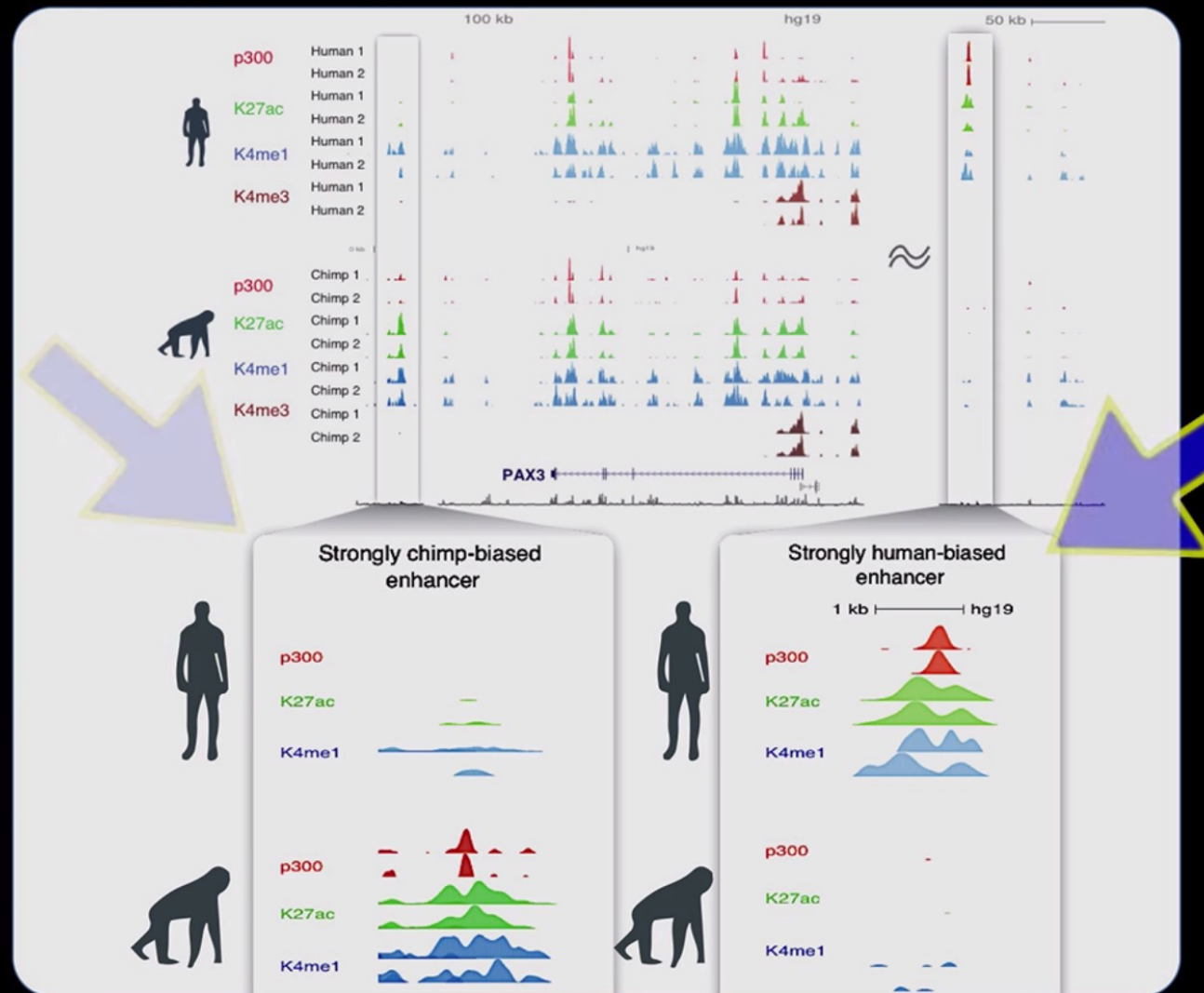




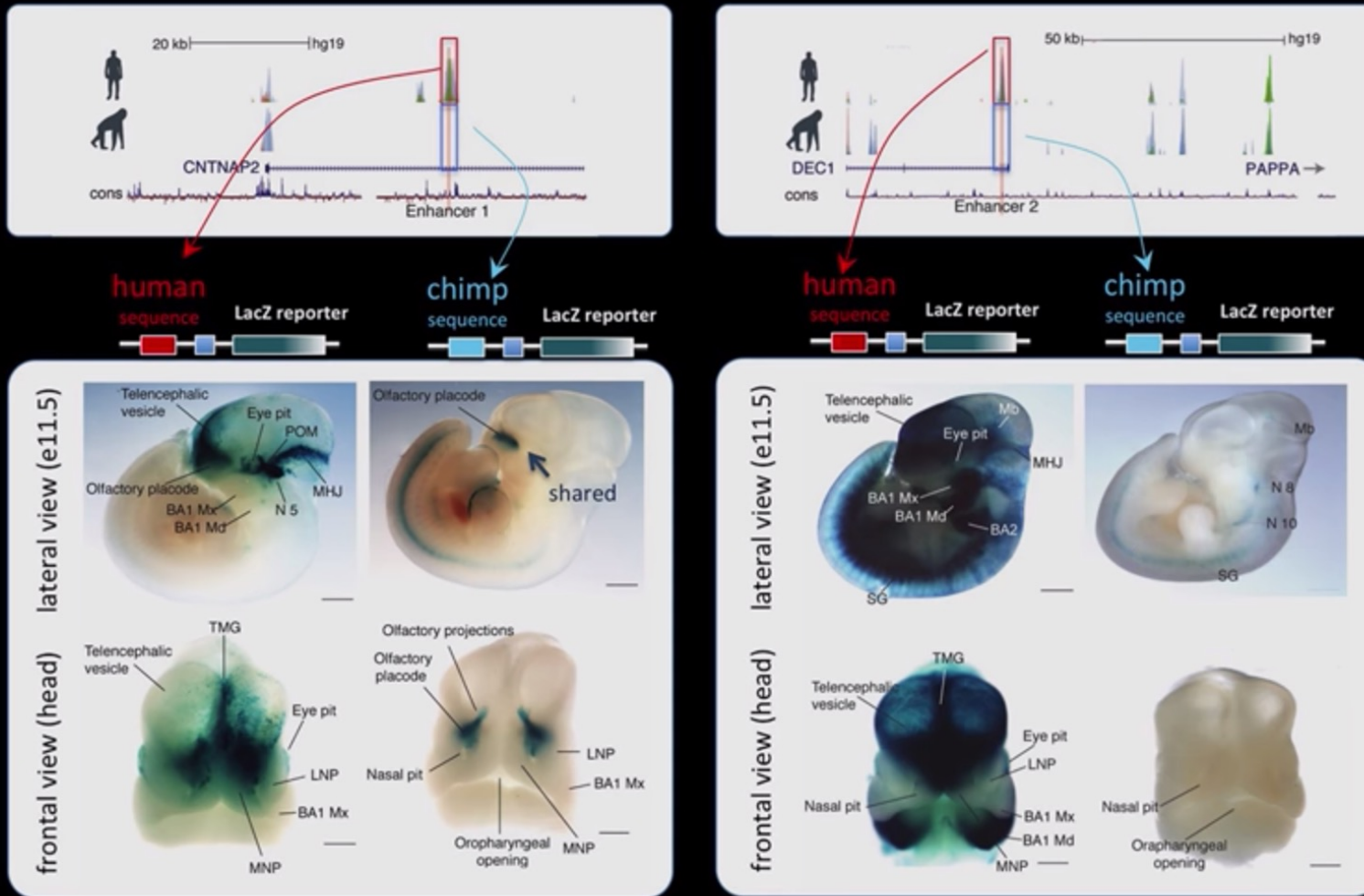
Lysine methylation is associated with transcription **activation** or **repression**



Discovery of species-biased enhancers

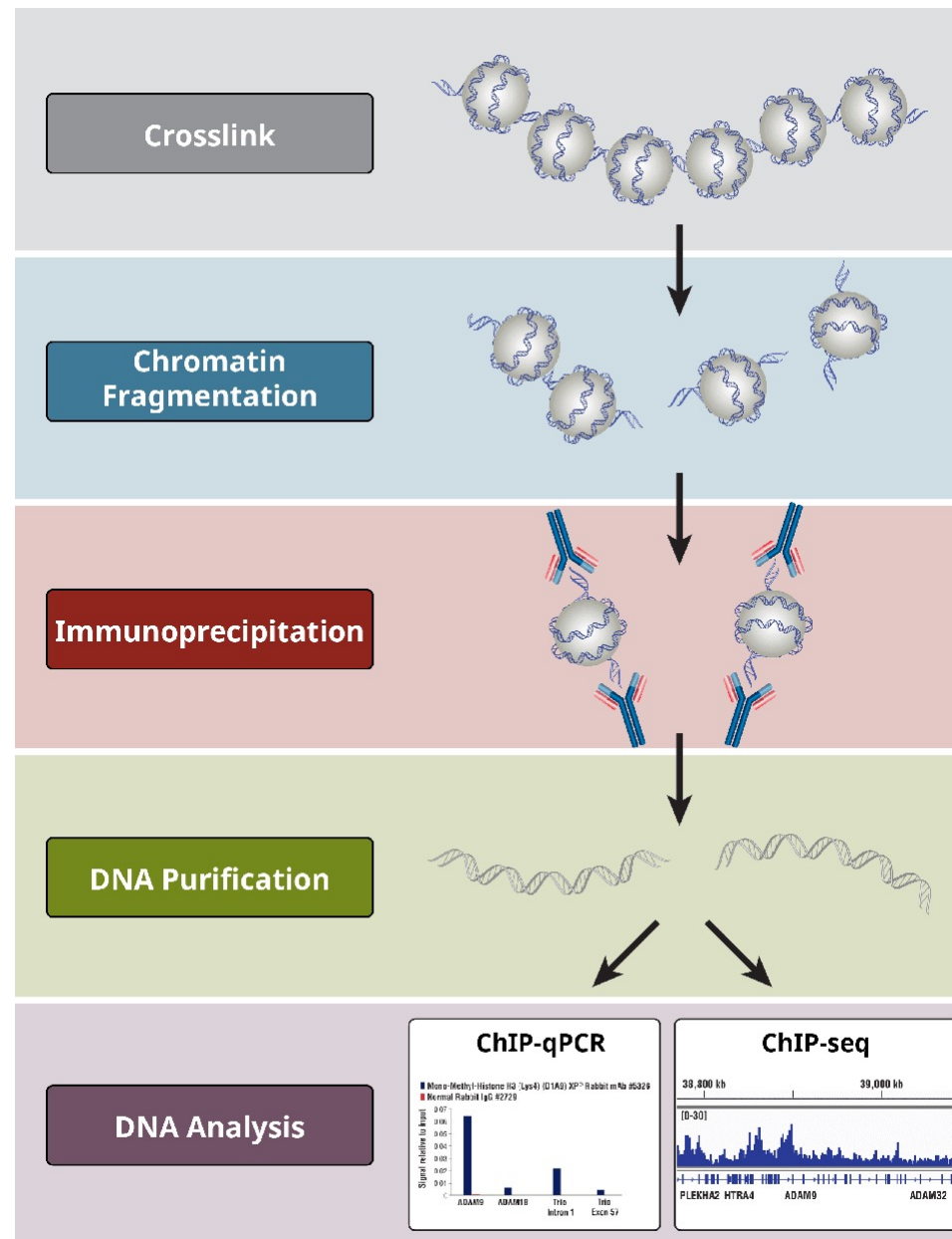


Species biased enhancers show divergent activity in head structures *in vivo*





ChIP-seq



Crosslink_ChIP-seq

“**Fixation**” refers to chemical means of killing cells and preserving them in a particular state and morphology.

There are two types of common fixatives:

- 1) Crosslinking fixatives — Fixatives that form covalent bonds with “methylene bridges.”

- Formaldehyde
- Glutaraldehyde



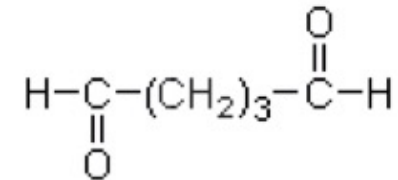
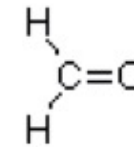
- Excellent morphological structure.
- But can lead to masking of antigen binding sites and increased autofluorescence.

- 2) Coagulating fixatives — Fixatives that precipitate proteins

- Methanol
- Acetone



- Immunoreactivity is well-preserved, and permeabilizes the sample.
- But morphology not as well preserved, and solubilizing of membranes can lead to loss of membrane dyes and membrane components.

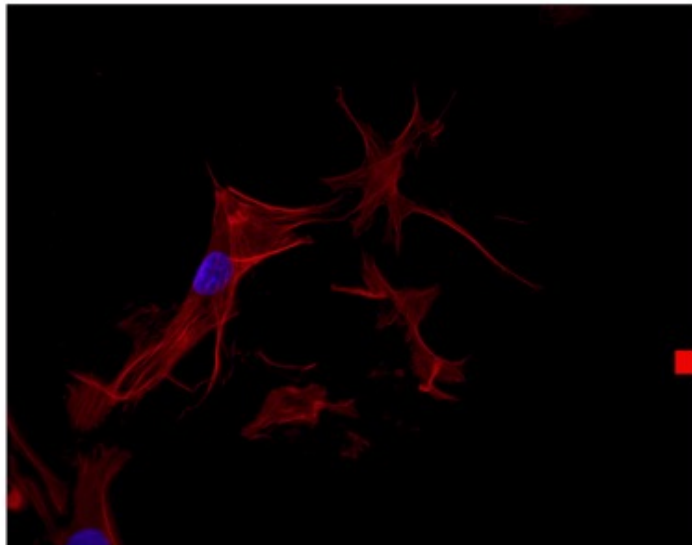


Crosslink _ChIP-seq

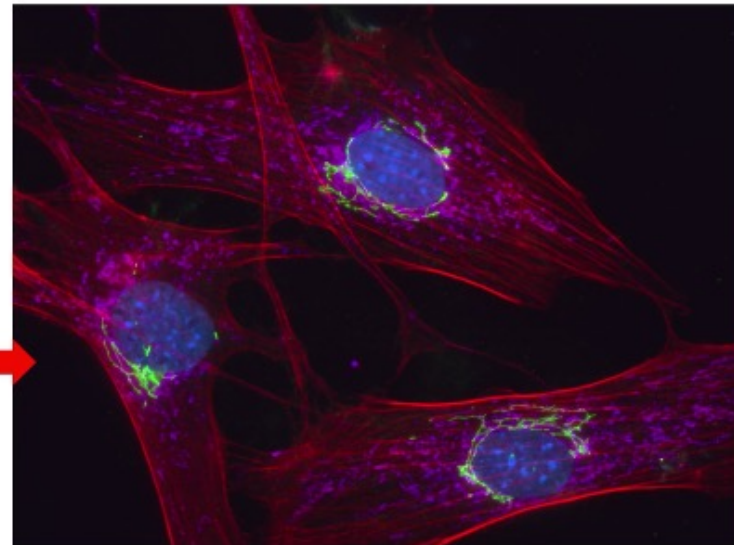
Proper handling during fixation can greatly affect the quality of staining:

- Always use freshly prepared high-quality formaldehyde.
- Use warm (37°C) 2–4% formaldehyde, incubated with sample for 10–15 minutes.
- You may dilute formaldehyde in complete media for some staining applications.

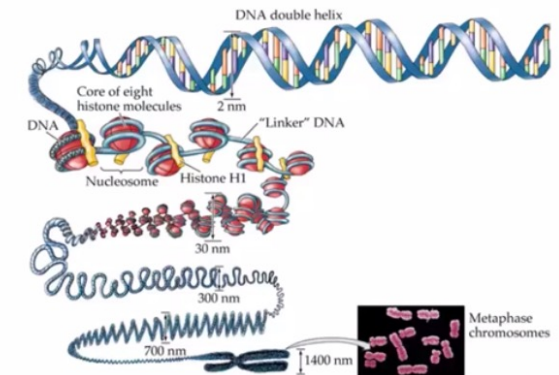
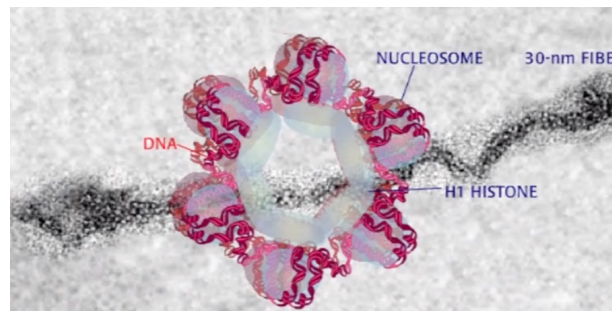
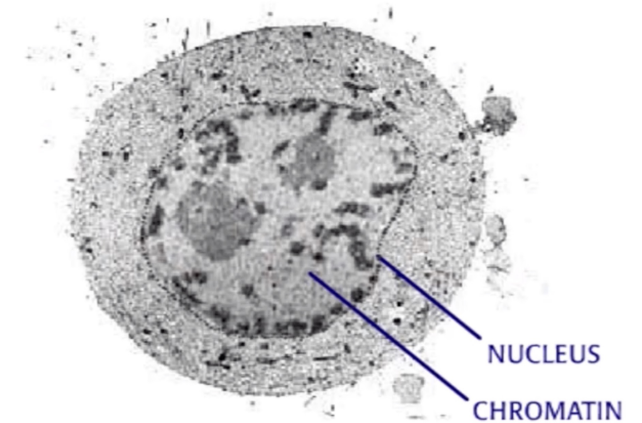
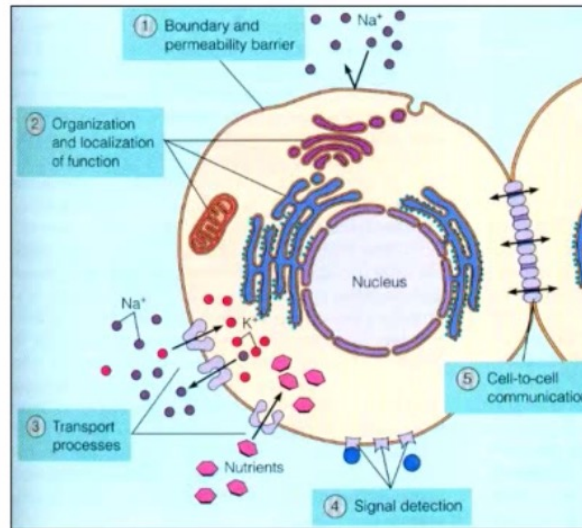
Poor Fixation



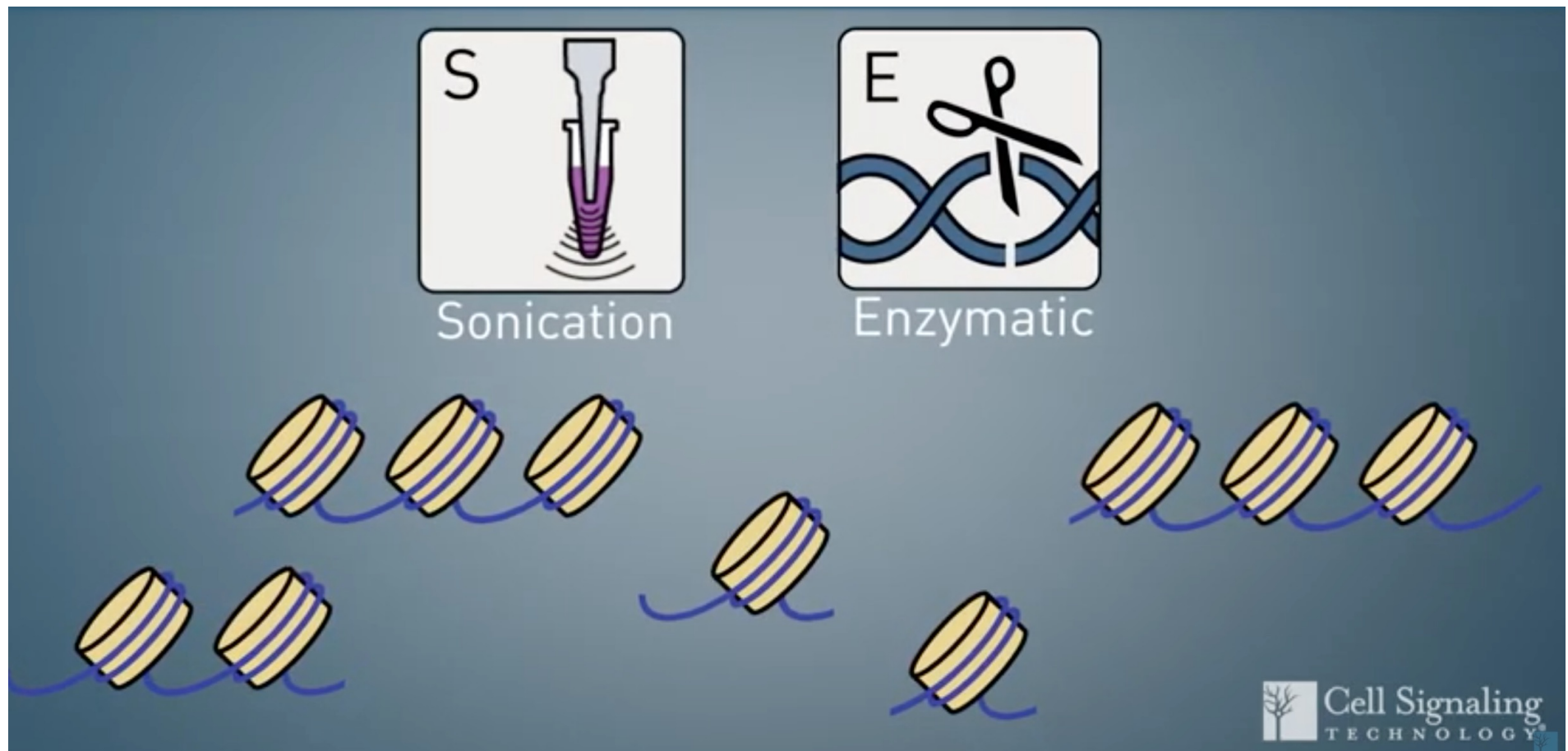
Proper Fixation



Cell lysis and nuclei lysis _ChIP-seq

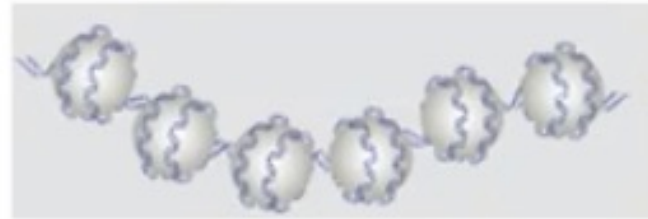


Chromatin fragmentation_ChIP-seq



Chromatin fragmentation_ChIP-seq

Chromatin Fragmentation



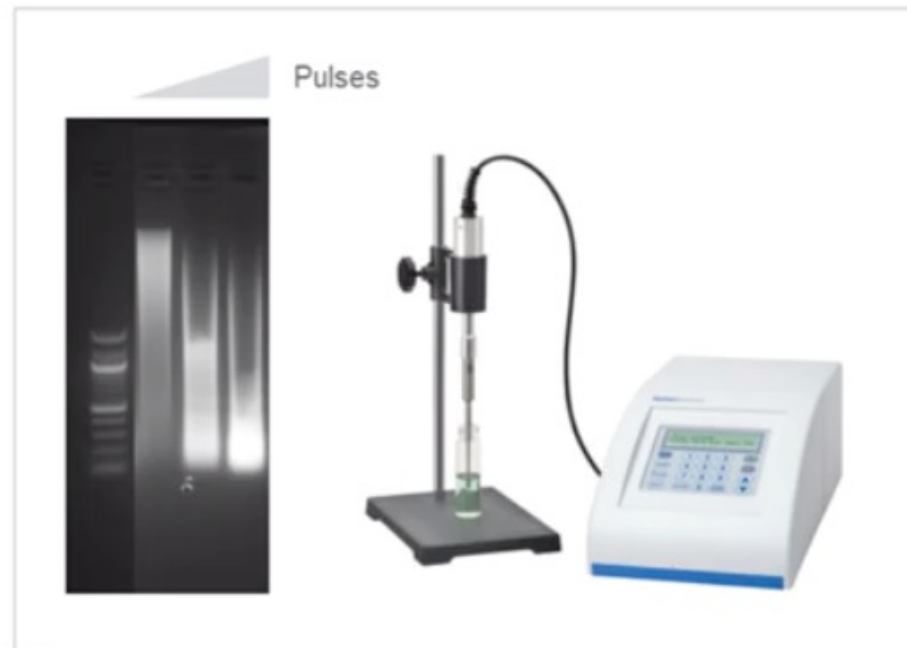
by sonication and/or nuclease/enzymatic digestion

fragment sizes between 200 and 800 base pairs

Chromatin fragmentation _ChIP-seq

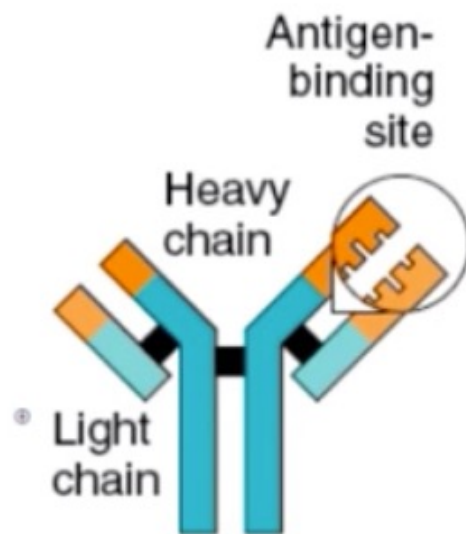
Mechanical method for fragmenting the chromatin

- Ideal range for chromatin is 200–800 bp
- Need to optimize for every cell line
- Need to optimize for every sonicator
- Keep samples very cold
- Prevent foaming
- Hands-on



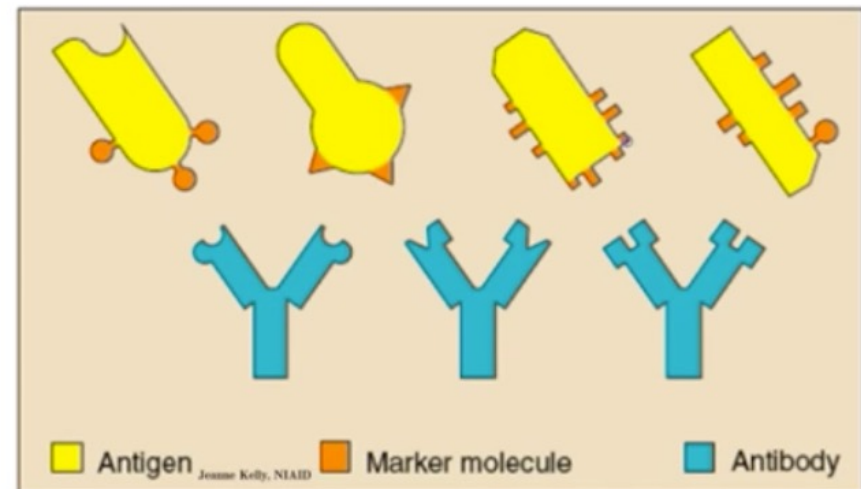
Immunoprecipitation _ChIP-seq

Antibody Structure

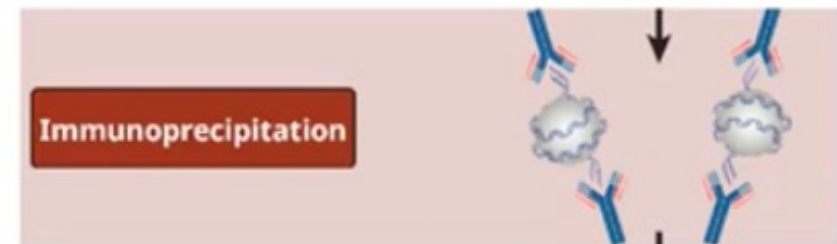
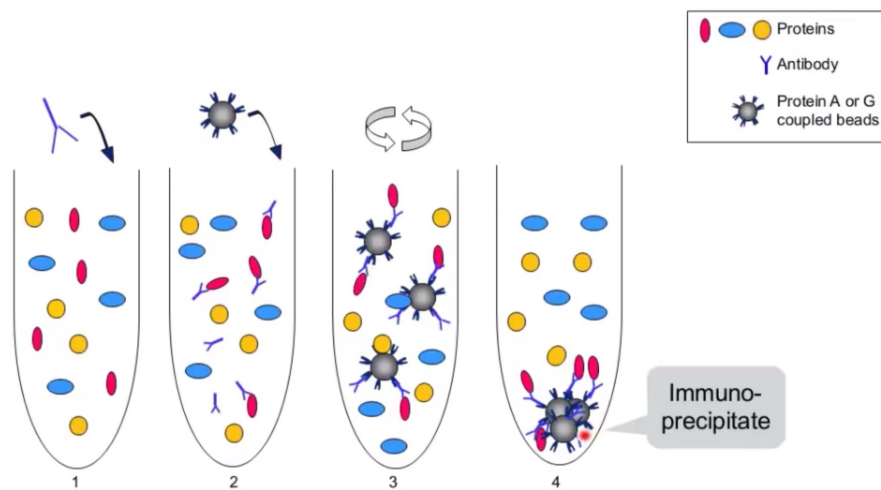


- Variable region
- Constant region

Antigen-Antibody Specificity



Immunoprecipitation _ChIP-seq



Capture and isolate DNA protein complex

DNA purification_ChIP-seq

Preparation of DNA

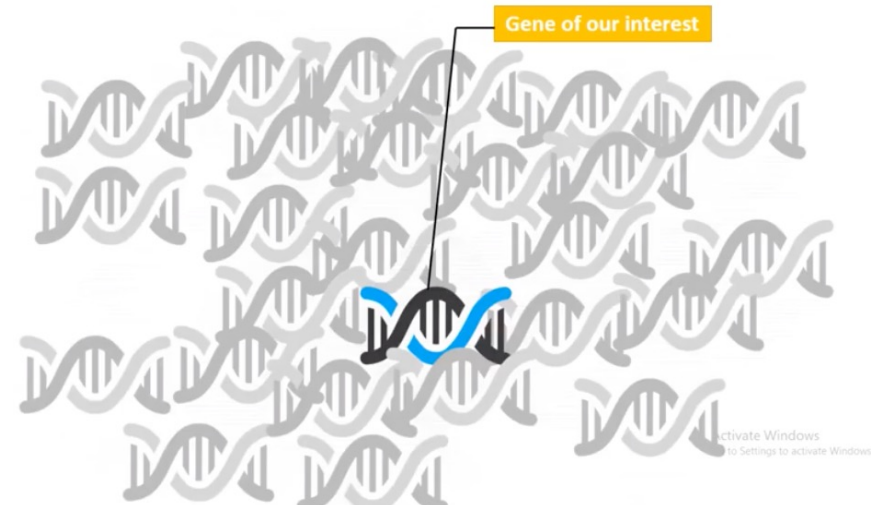
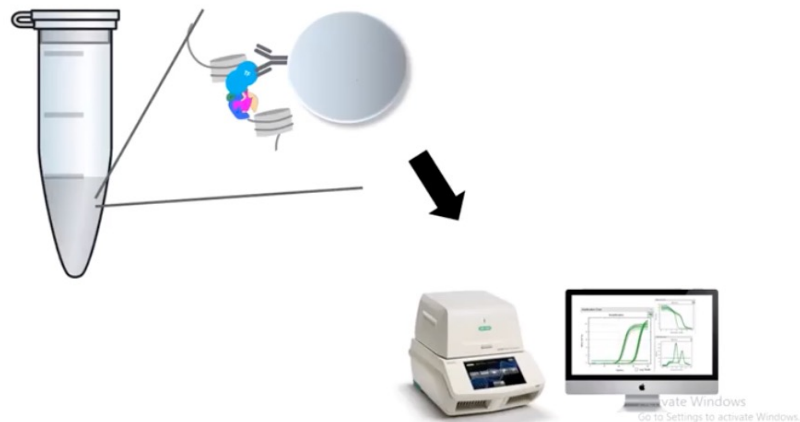
Preparation of DNA

Cross-link reversal
and DNA purification

DNA Purification



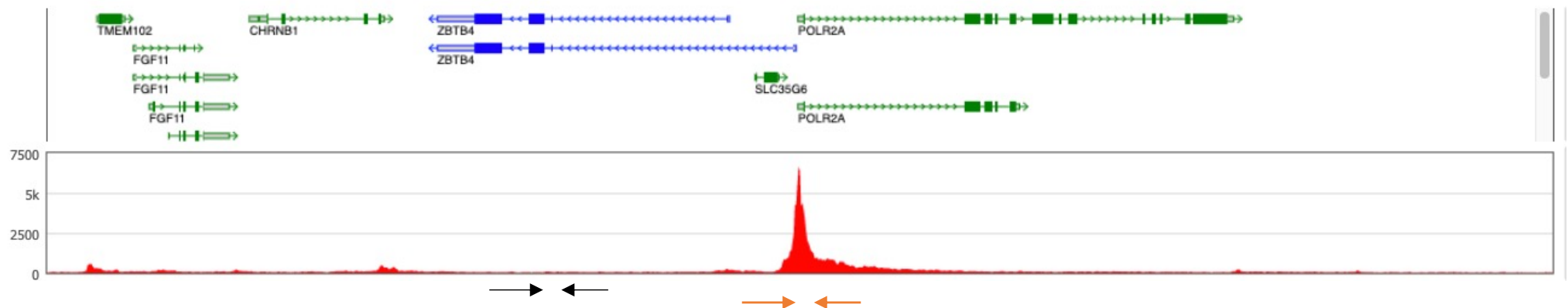
Enrichment QC _ChIP-seq



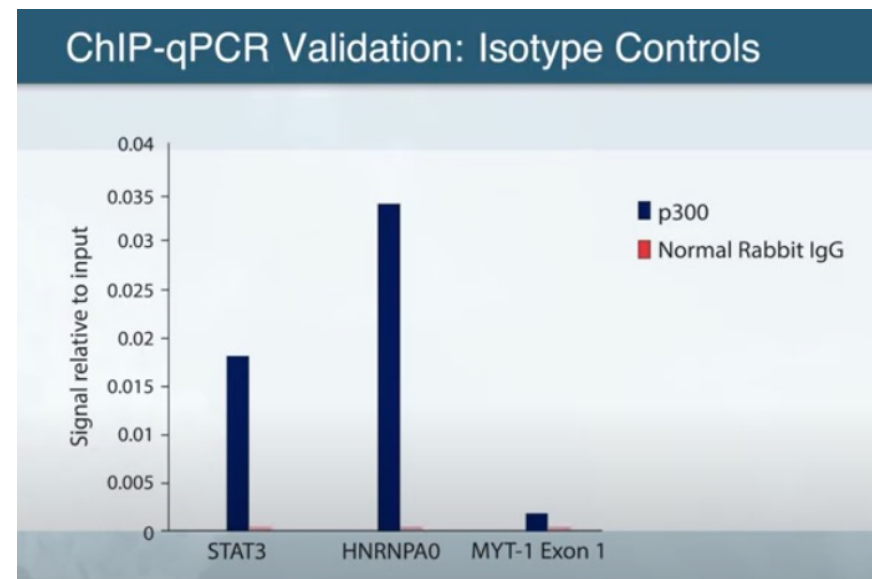
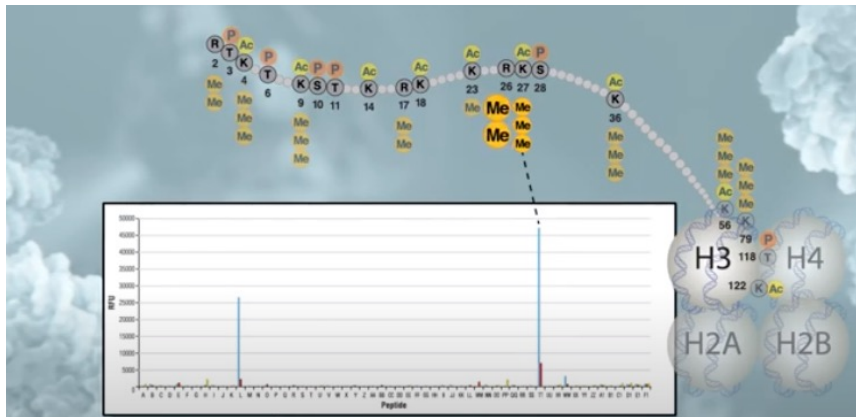
ChIP QC



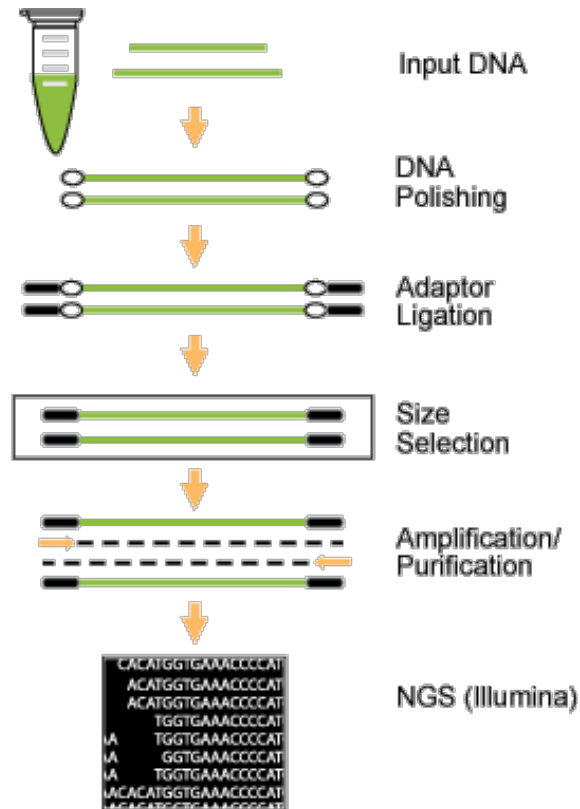
hg19, chr17:7335646-7440093, 104 kb



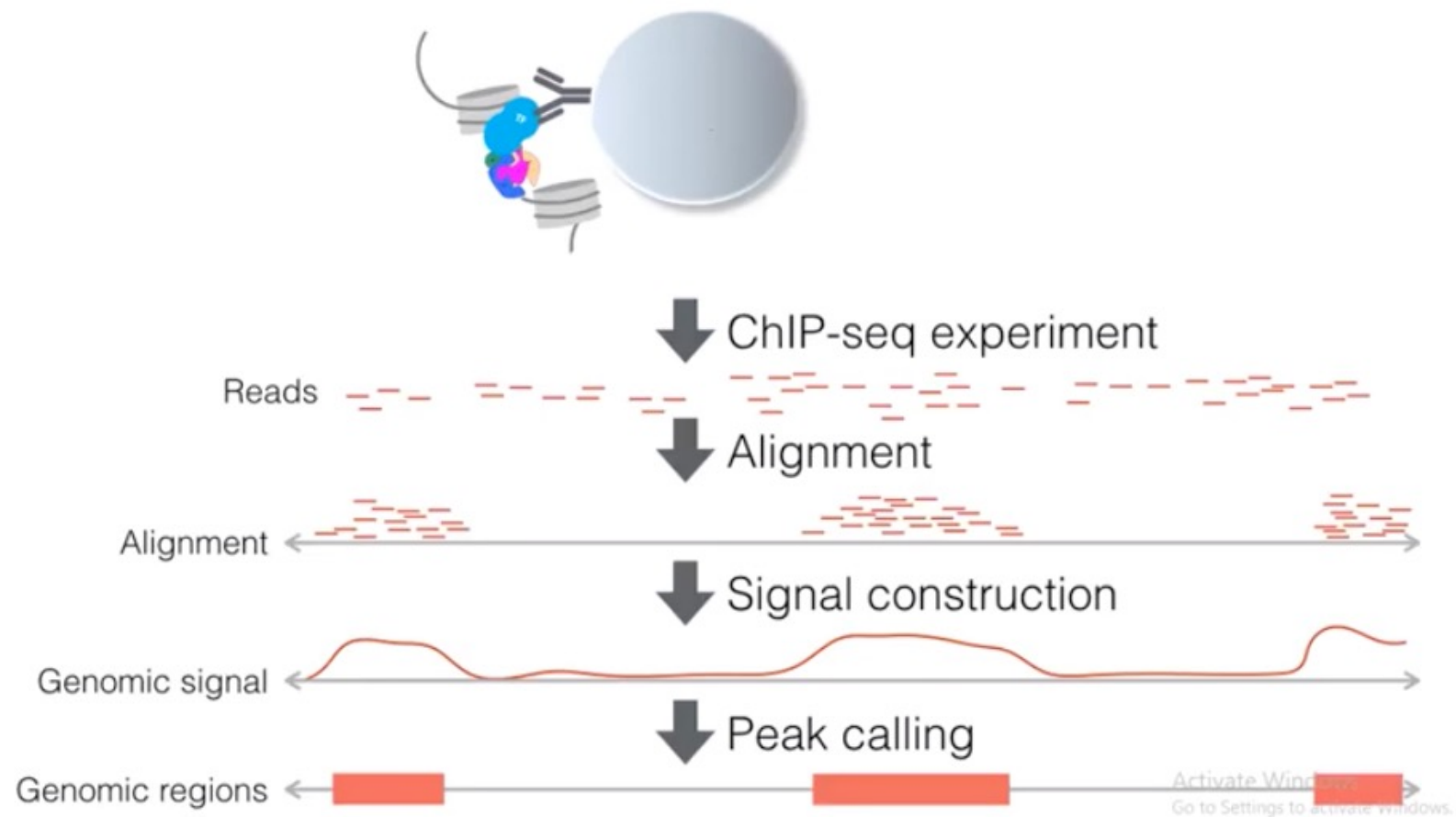
Enrichment QC _ChIP-seq



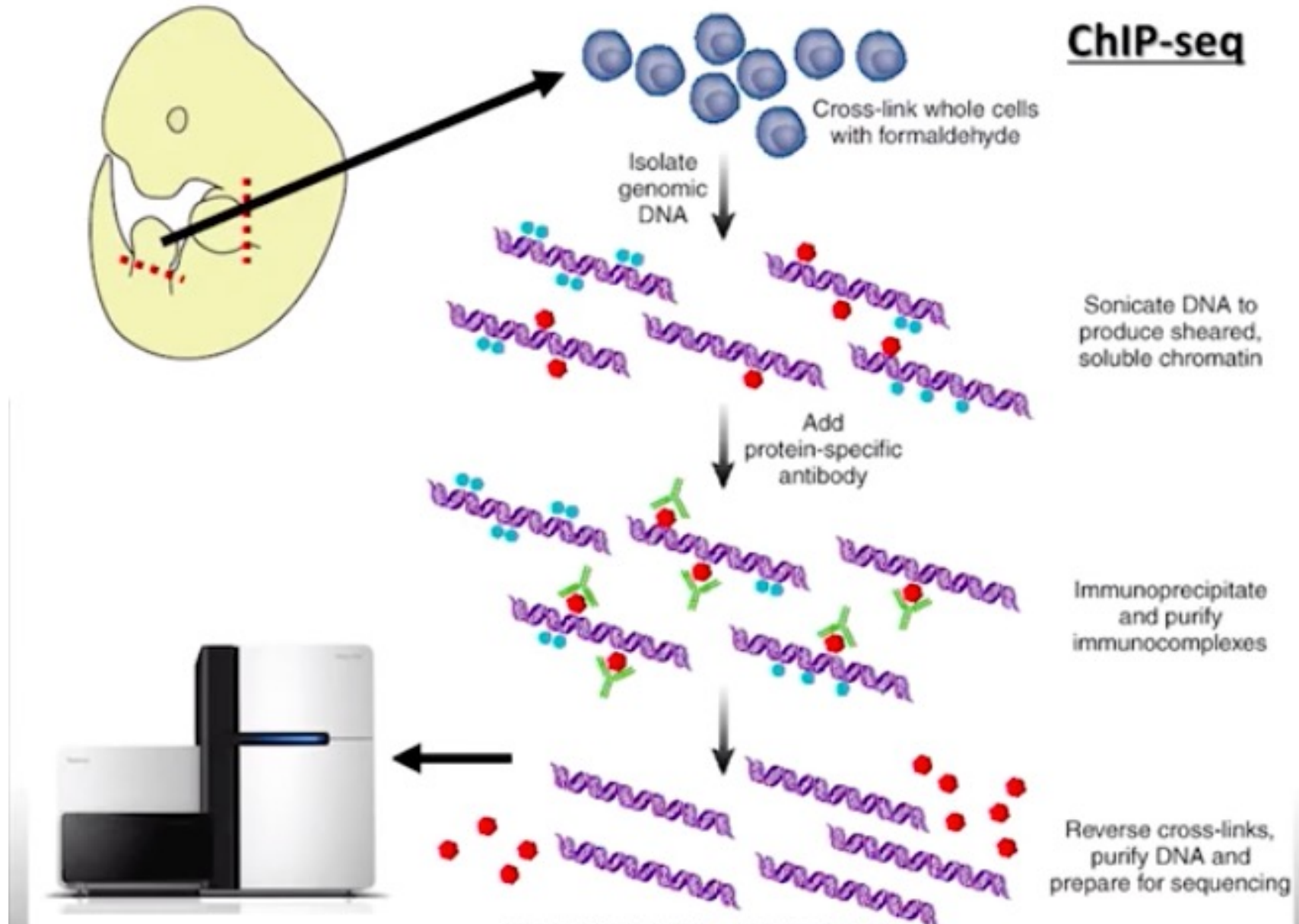
DNA library generation and sequencing _ChIP-seq



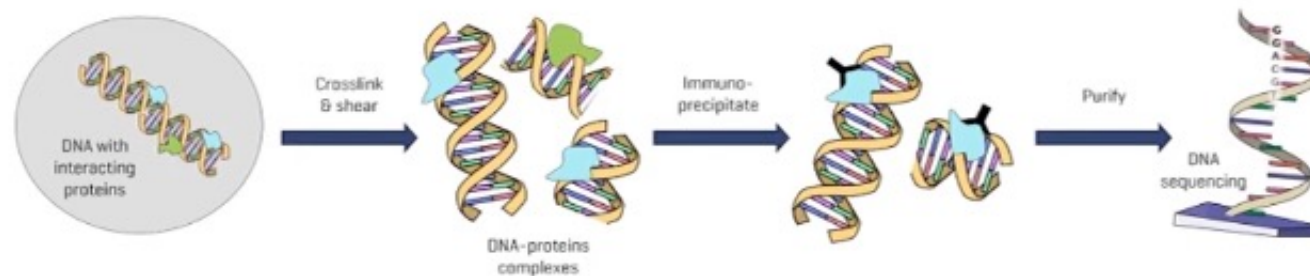
Results_ChIP-seq



ChIP-seq



Chromatin Immunoprecipitation (ChIP)



"ChIP-seq" ~55,000 articles (Google Scholar)

Limitations

Low efficiency (millions of cells)

High background (tens of millions of reads)

Low resolution (hundreds of bp)

Most antibodies fail

Chromatin profiling by enzyme tethering

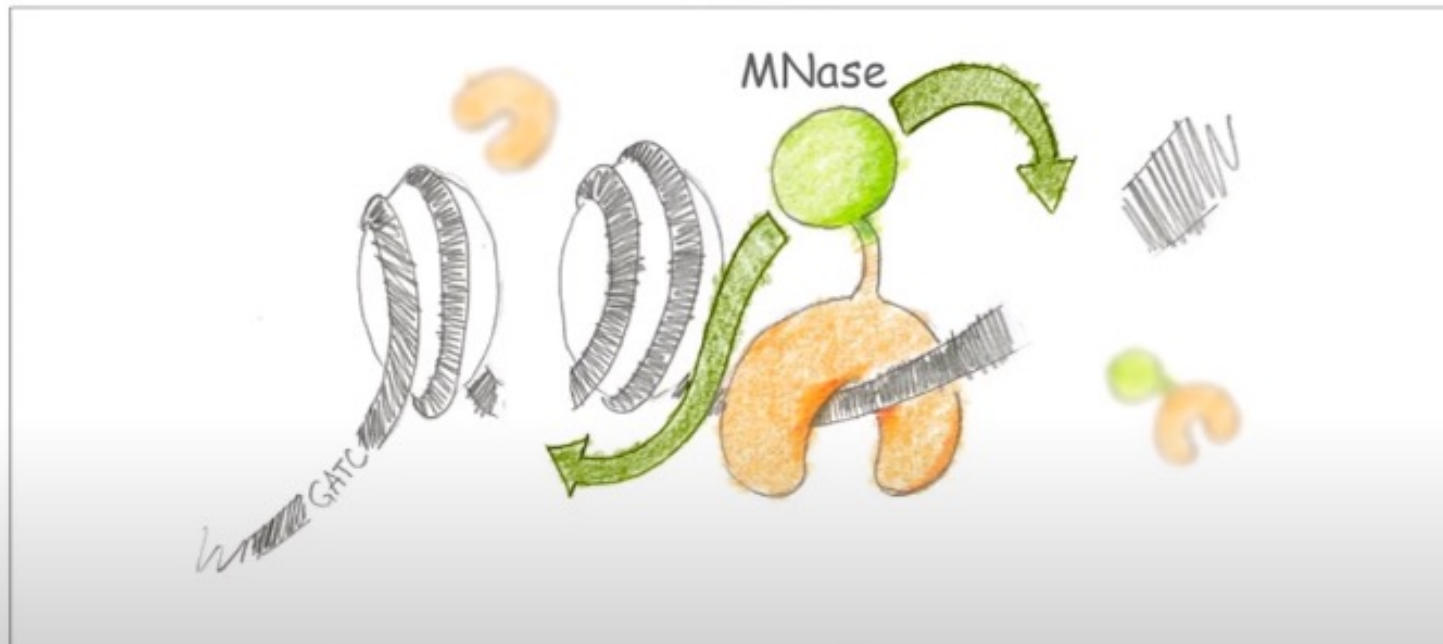
DamID



DamID – van Steensel & Henikoff (2000) *Nat Biotech*

Chromatin profiling by enzyme tethering

ChEC



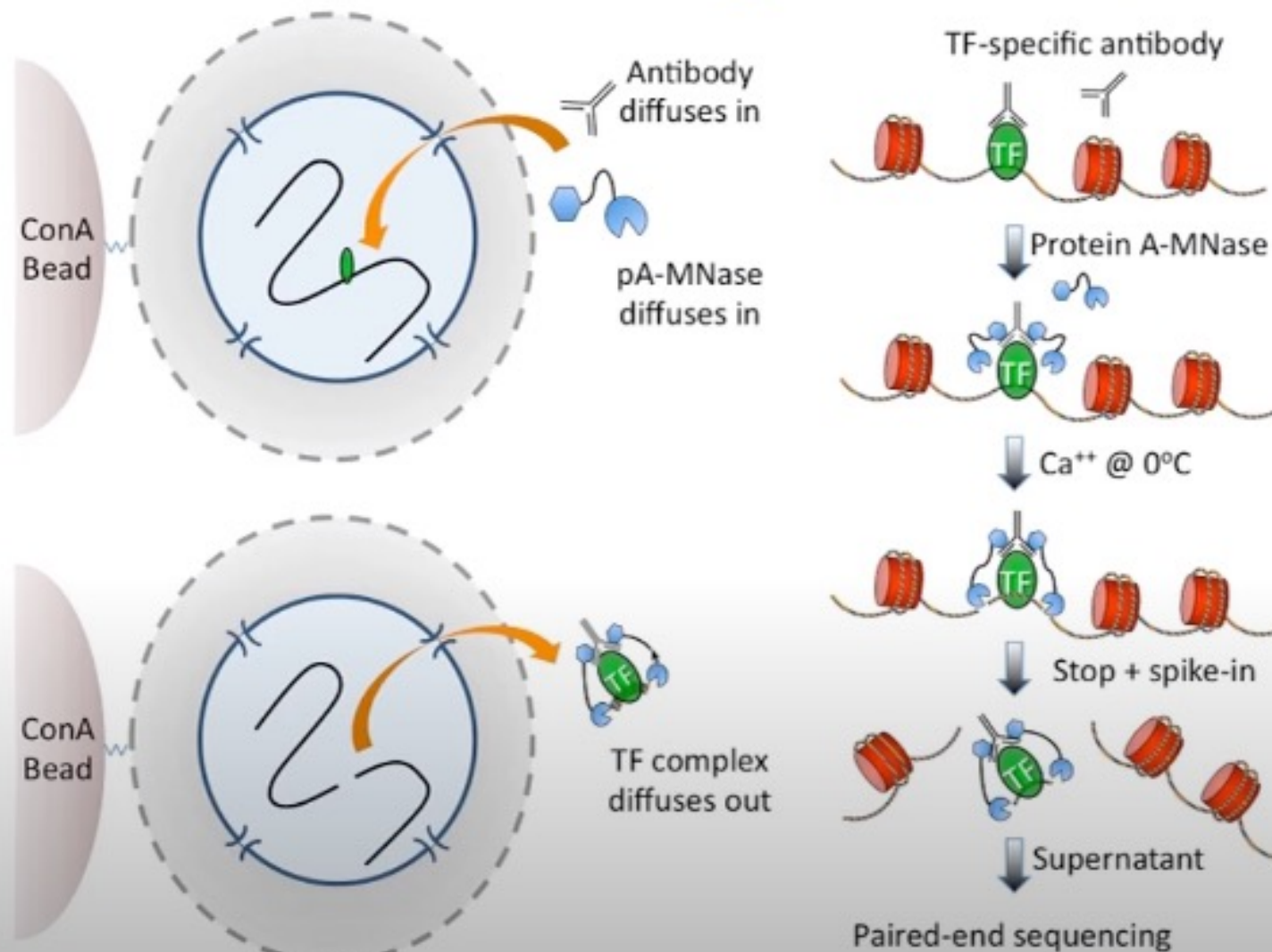
ChIC and ChEC – Schmid *et al.* (2004) *Mol Cell*

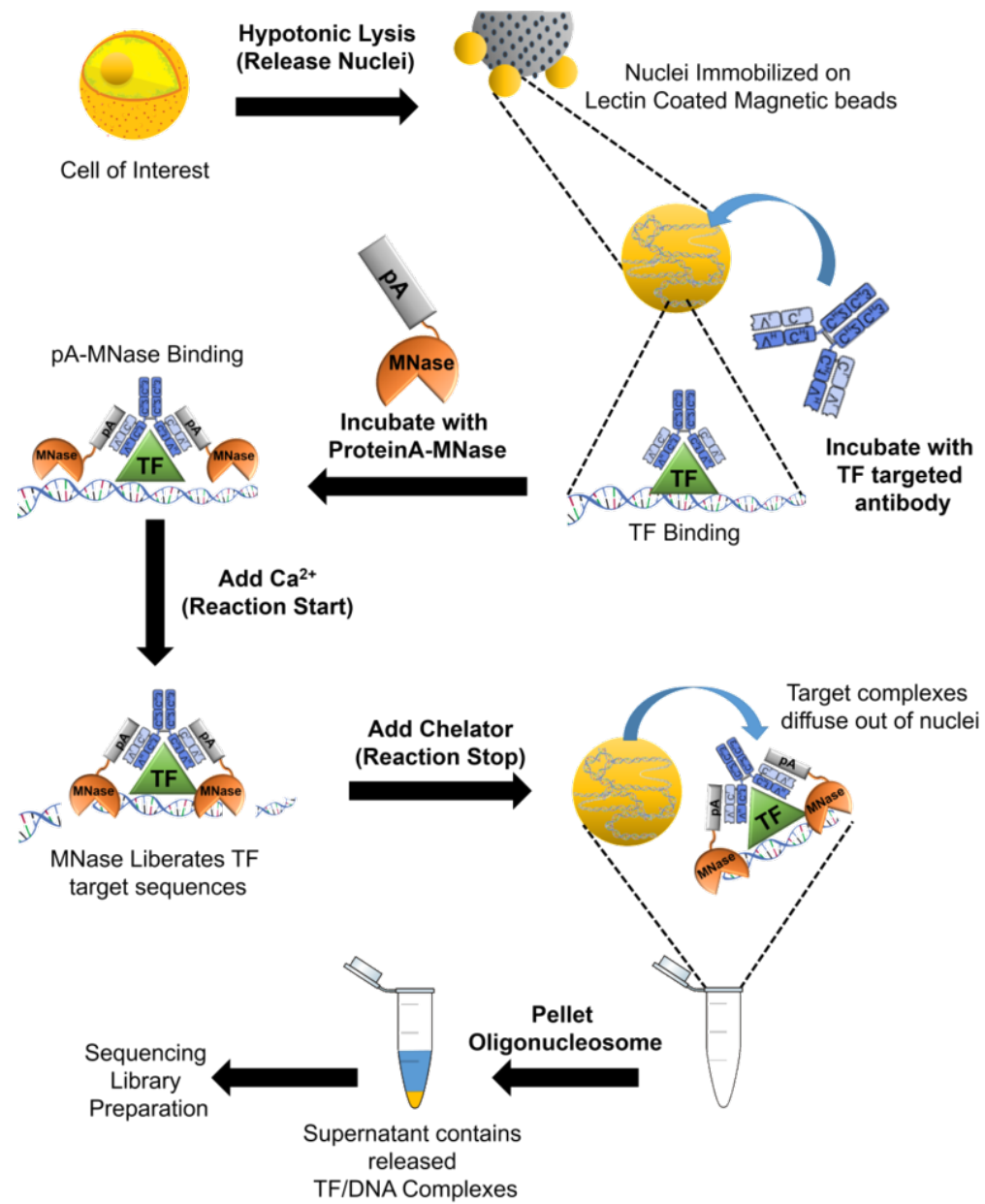
ChEC-Seq – Zentner *et al.* (2015) *Nat Commun*



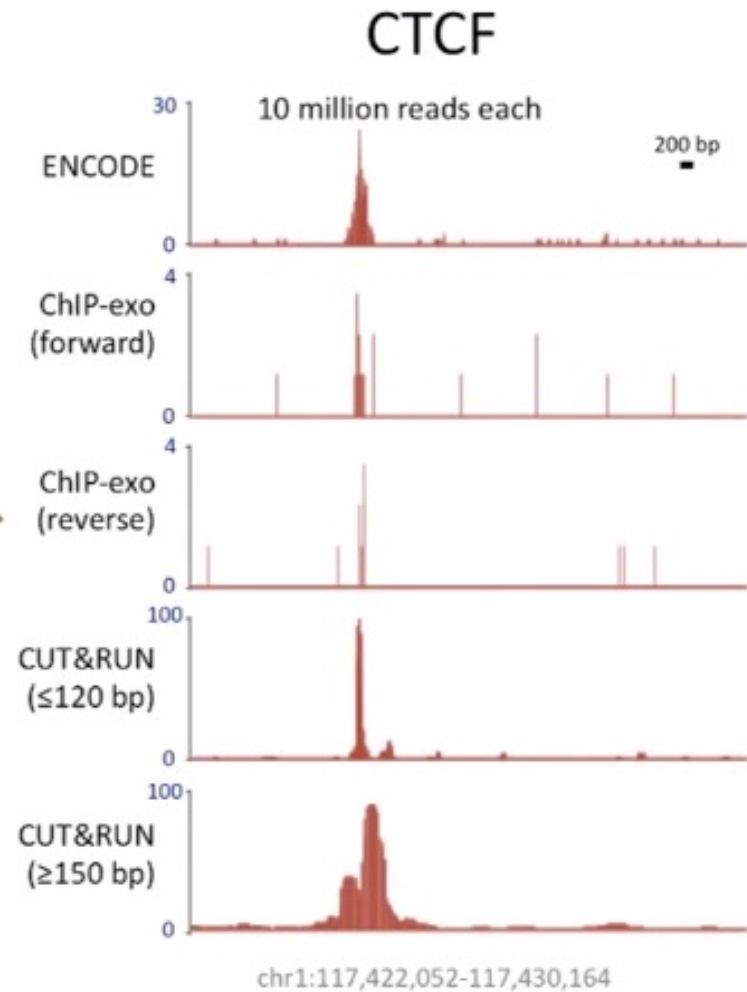
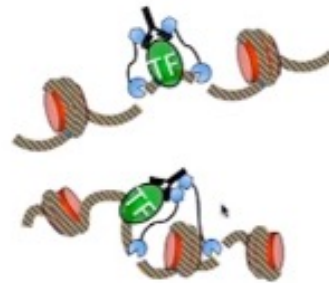
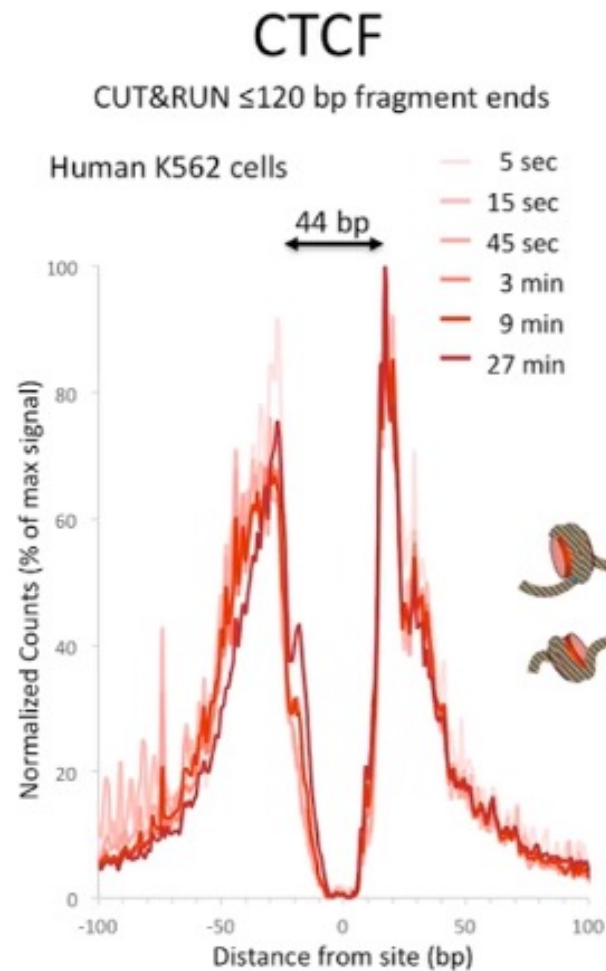
CUT&RUN

CUT&RUN (Cleavage Under Targets & Release Using Nuclease)



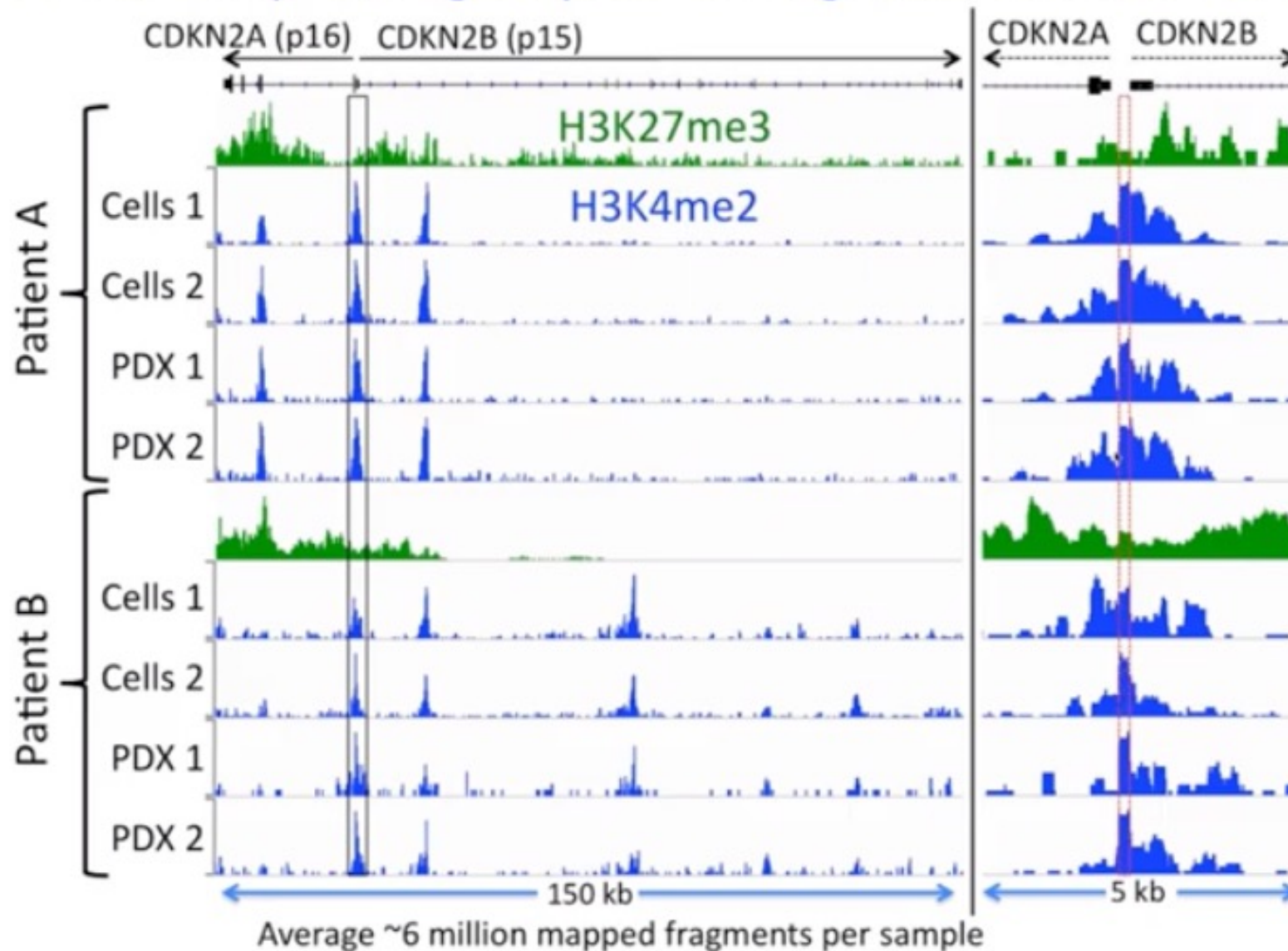


CUT&RUN for base-pair resolution maps

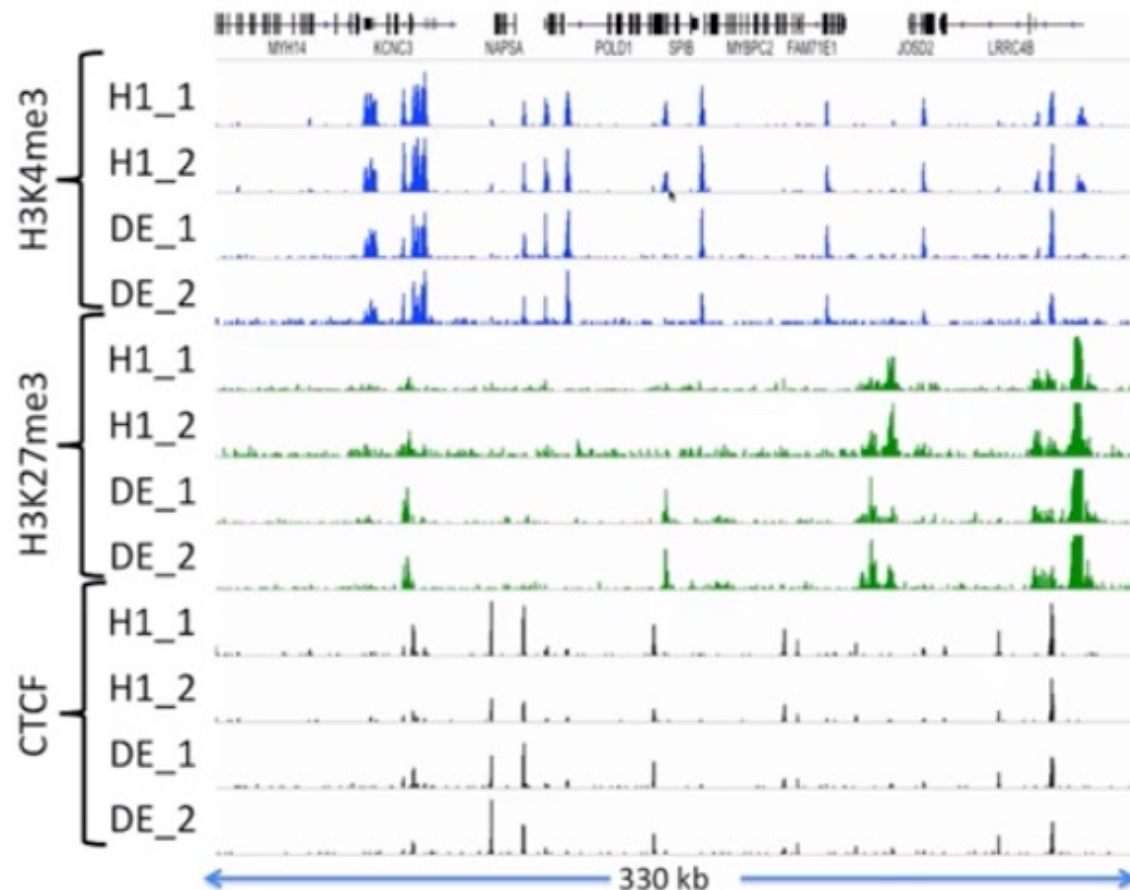


Skene & Henikoff (2017) *eLife*

CUT&RUN profiling of pediatric glioma-derived PDXs

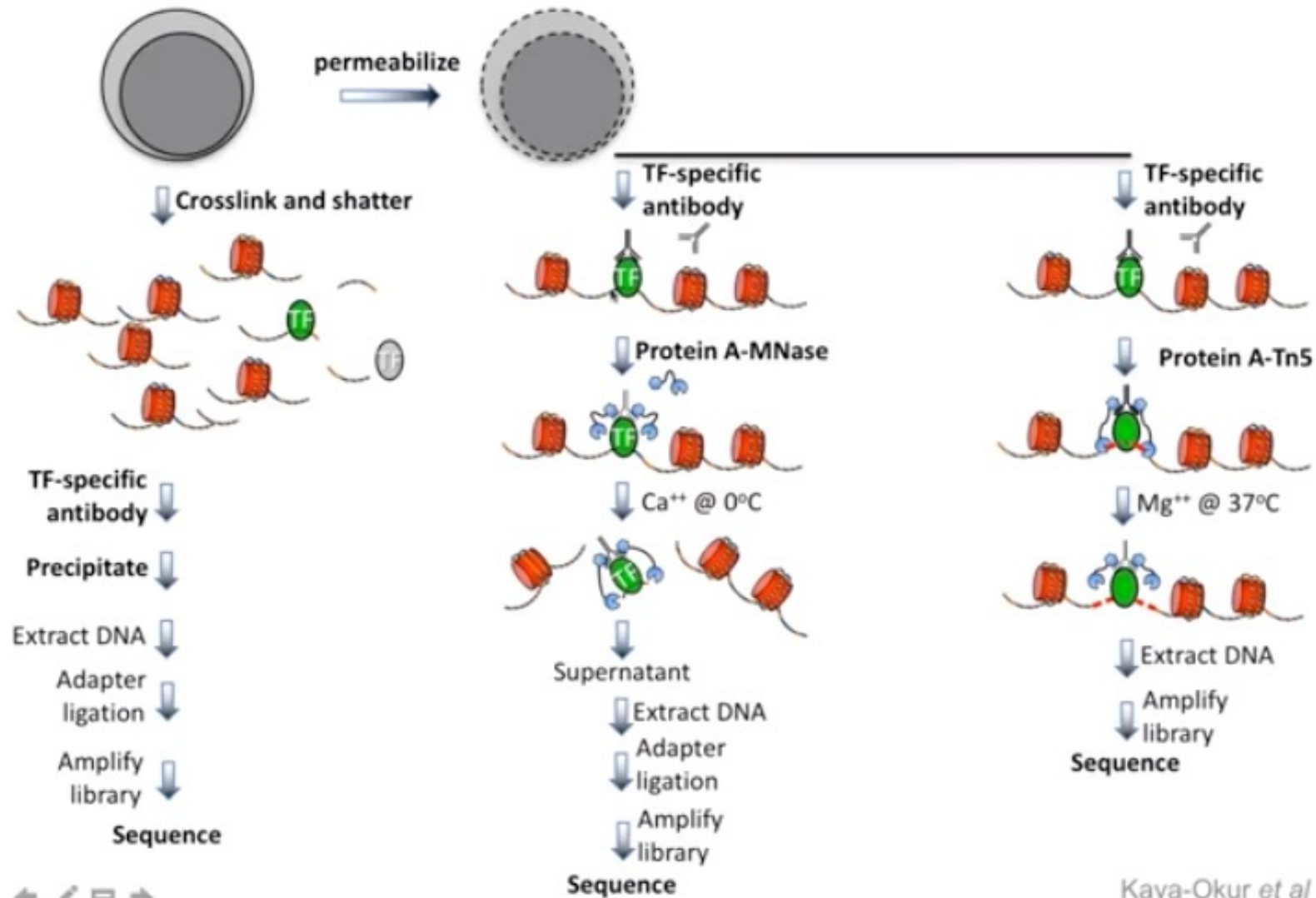


CUT&RUN profiling of differentiating cells



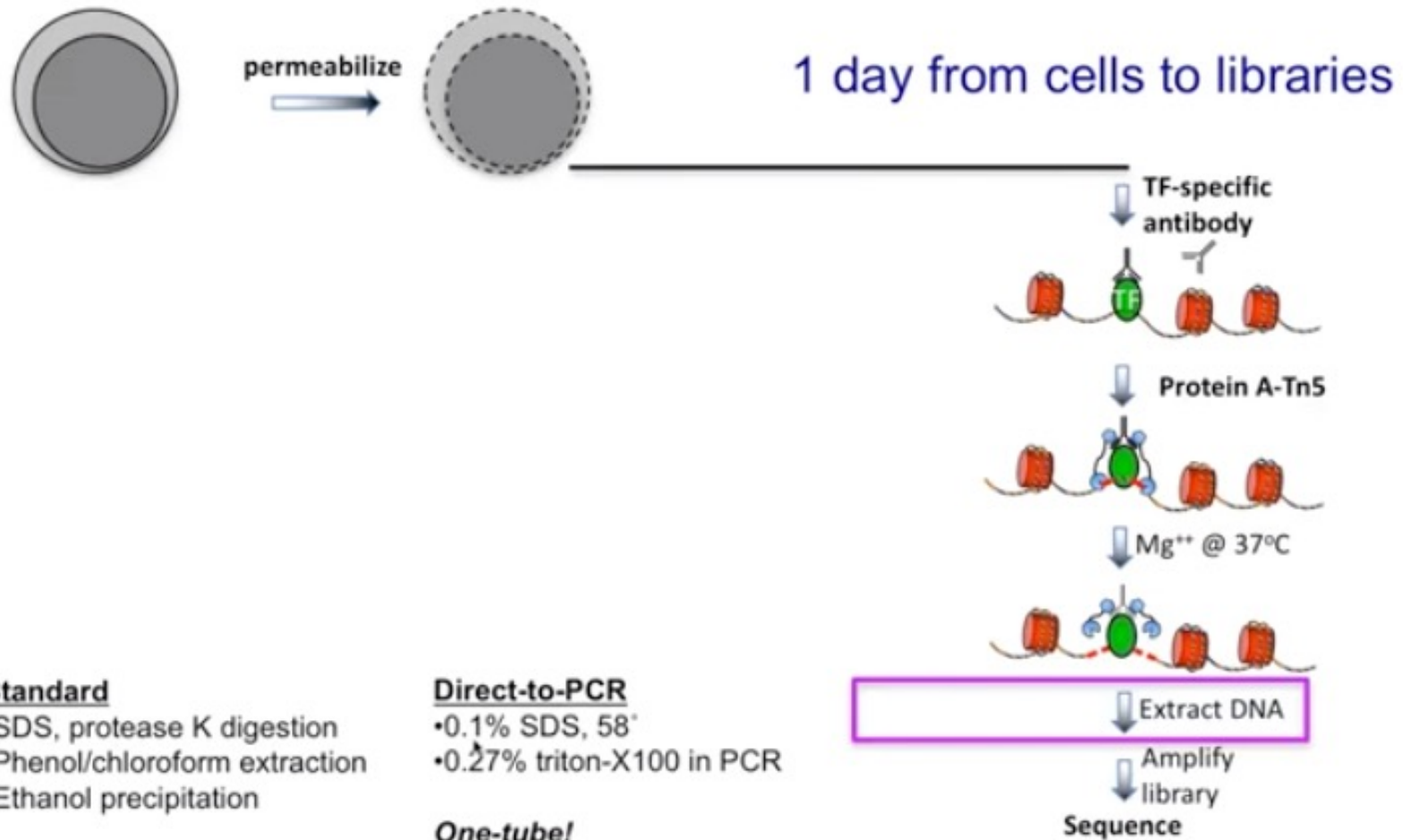
Vedio2_Chapter11_ChIP-seq+vs+CUT&RUN+
+Learn+the+benefits+of+CUT&RUN+sequencing+for+enhanced+chromatin+profiling.mp4

Improved chromatin profiling by CUT&Tag



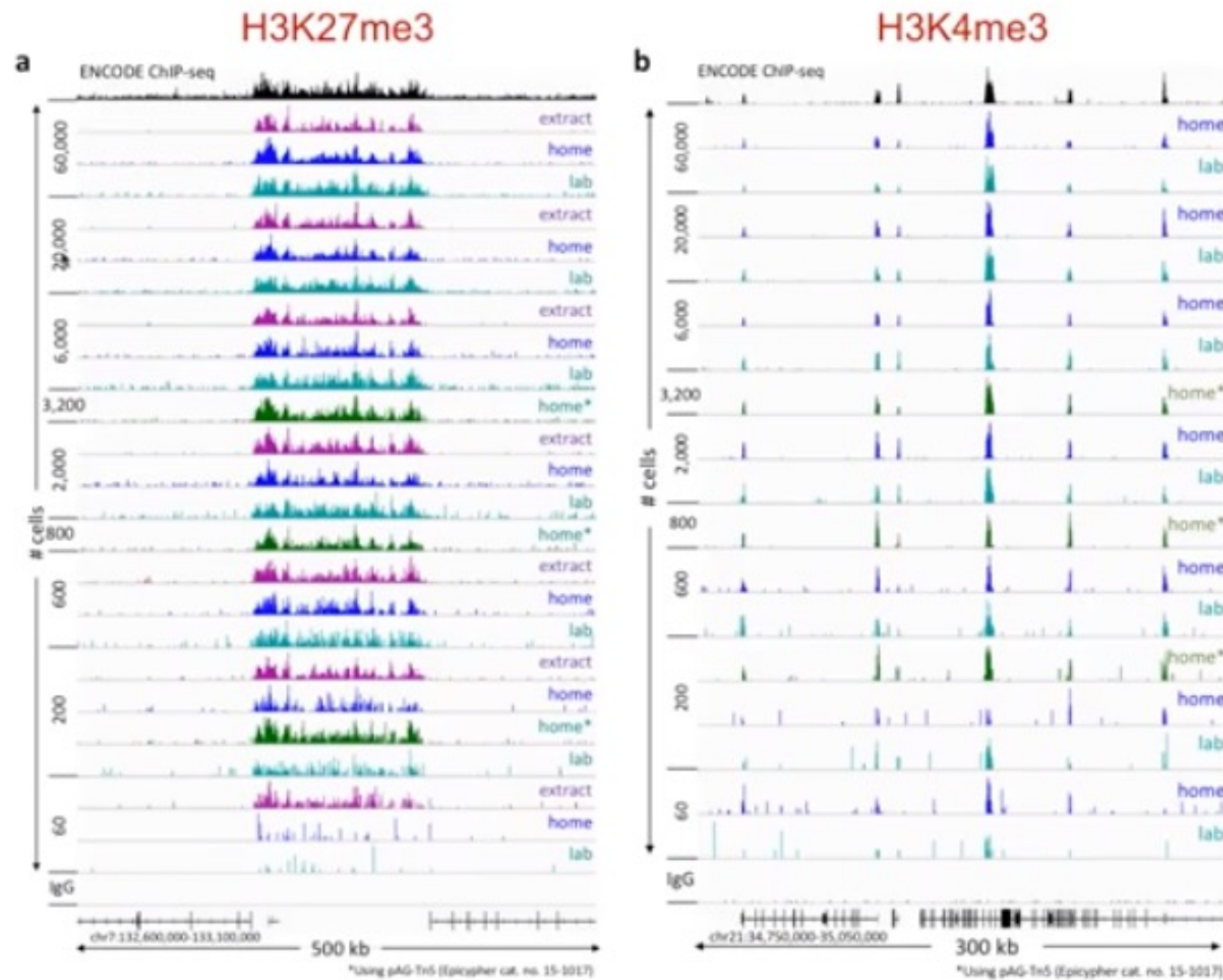
Kaya-Okur *et al* (2019)

Improved chromatin profiling by CUT&Tag



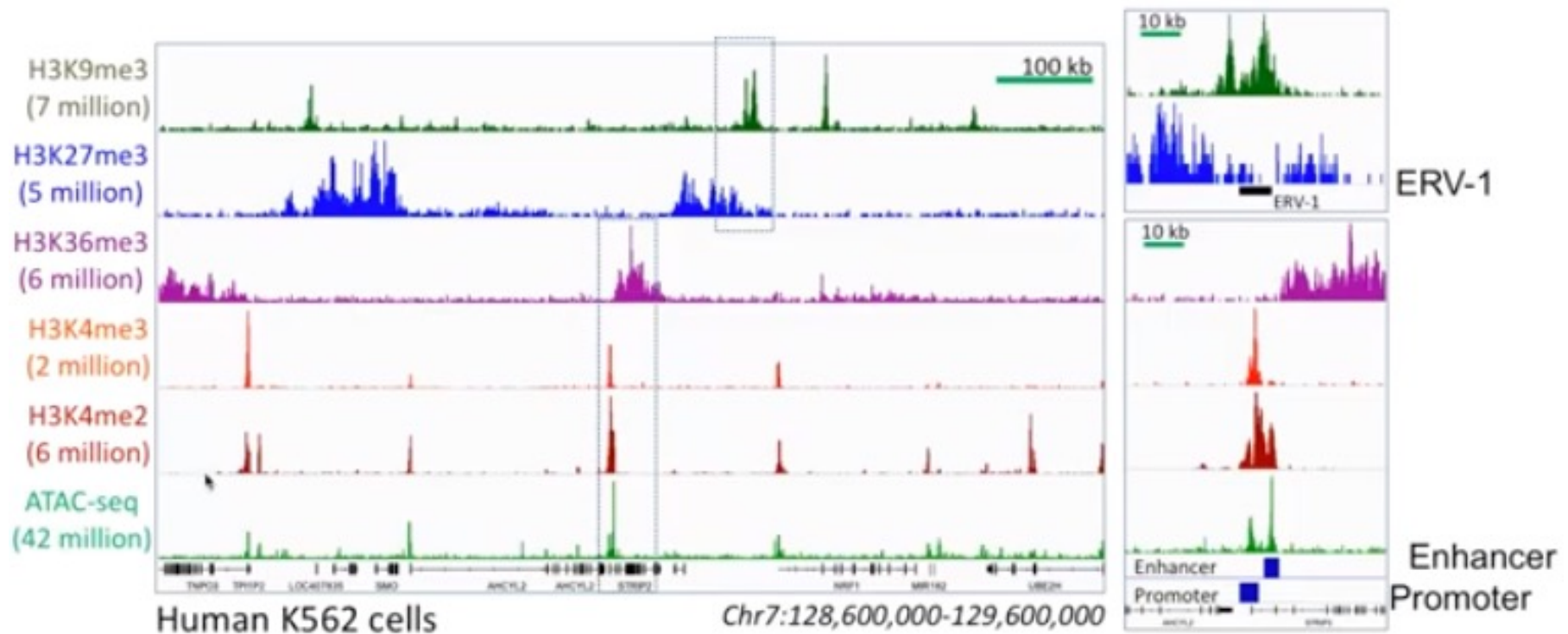
Kaya-Okur et al (2019)

Data quality with CUT&Tag@home



A simple view of the regulated genome

H3 N-tail: ⁴ART⁹KQTARKSTGGKAPRKQLATKAARK²⁷SAPATGGV³⁶KKPHR

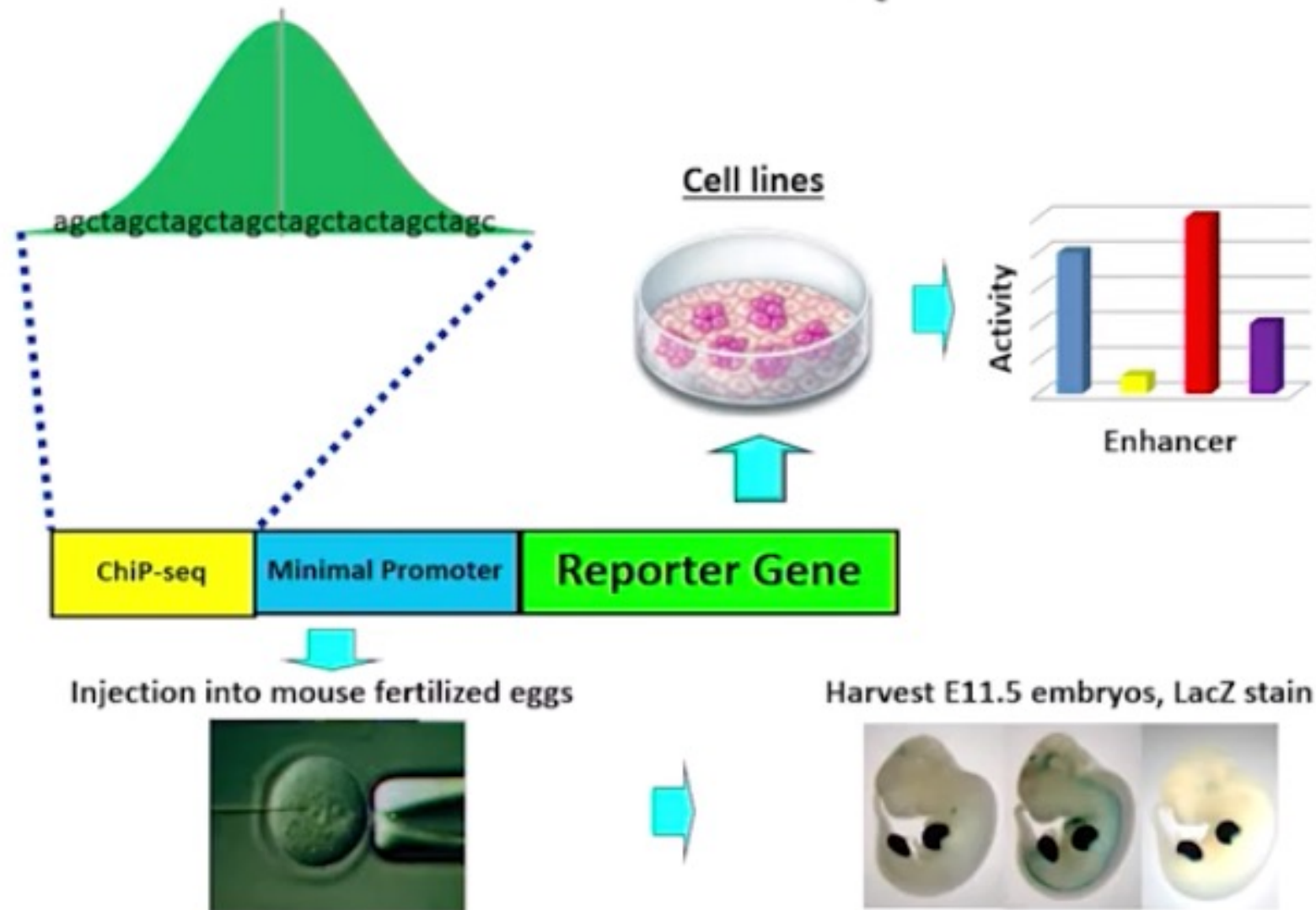


CUT&Tag@home segments the human epigenome with five H3 lysine methylation marks: Screenshot of a representative 1 Mb region of the human genome.

Vedio3_Chapter11_CUTANA™+CUT&Tag+Technology+Ultra-sensitive+Chromatin+Profiling.mp4

Enhancer assays

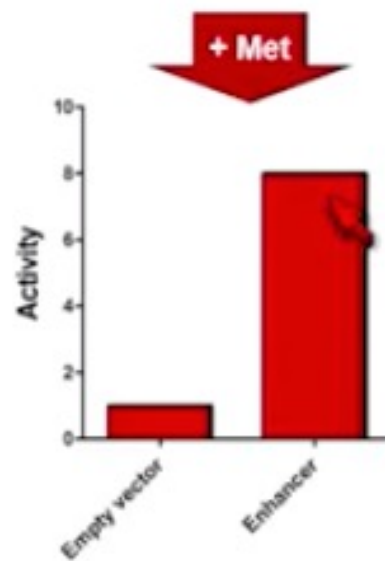
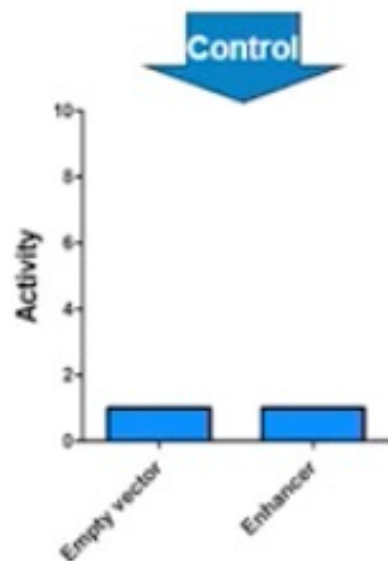
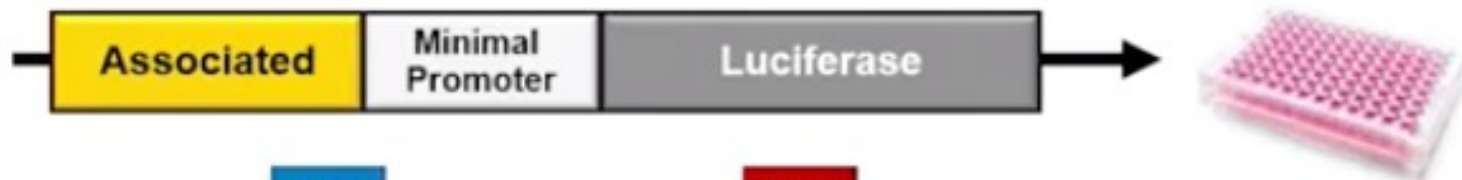
Enhancer Assay



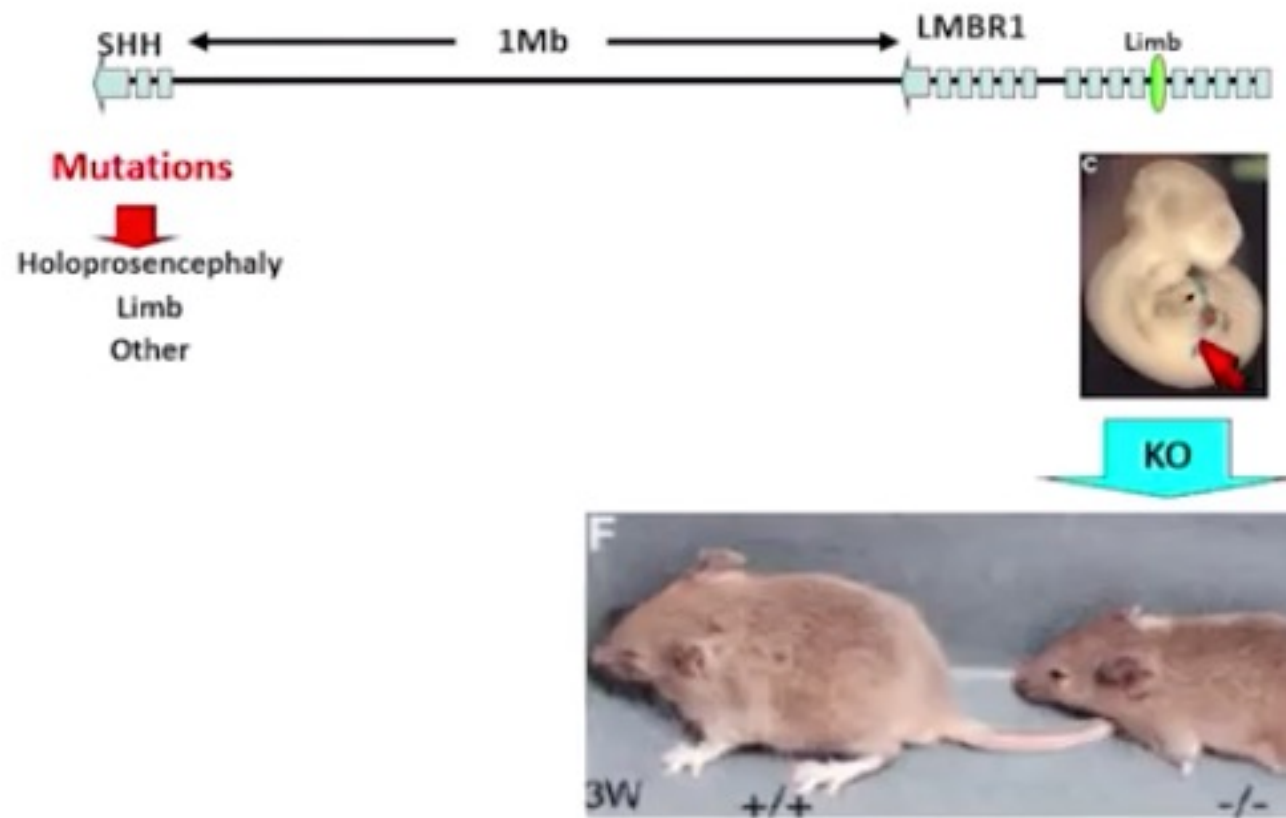
Functional testing of rifampin-dependent regions



Difference?

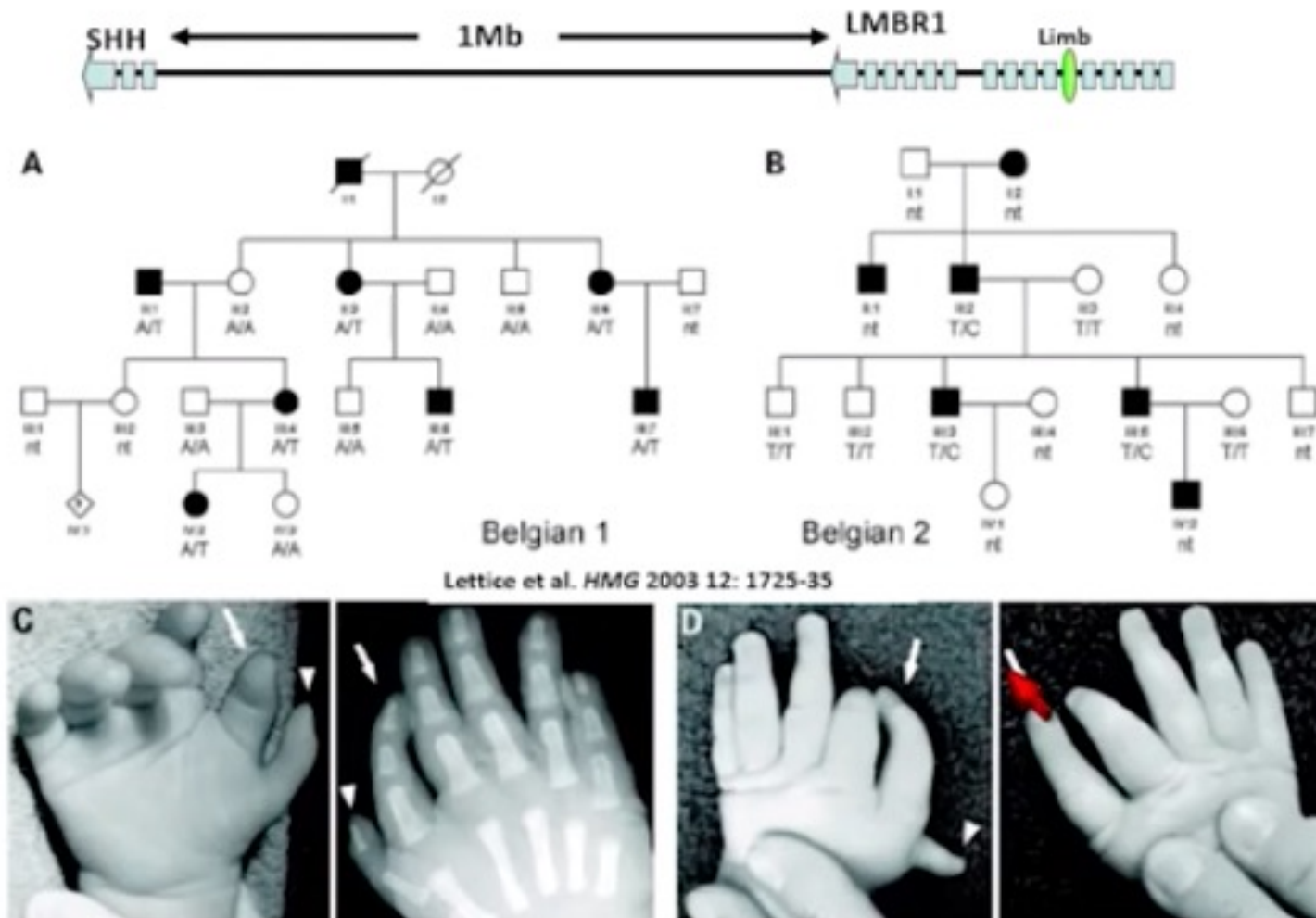


Mutations in the SHH limb enhancer (ZRS) lead to limb malformations



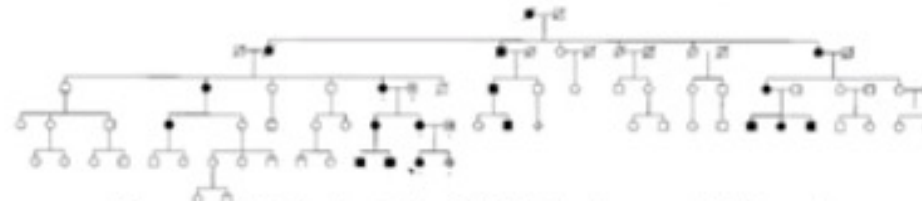
Sagai et al. *Development* 2005 132: 797-803

Mutations in the SHH limb enhancer (ZRS) lead to limb malformations



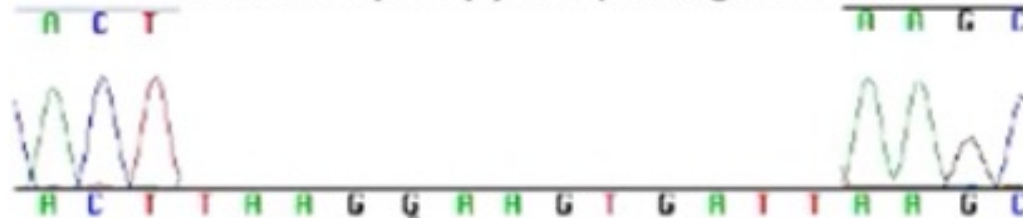


13bp insertion in the *SHH* limb enhancer

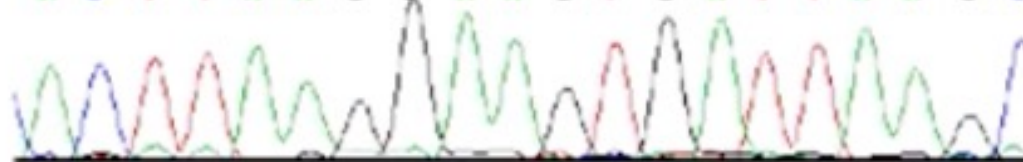


Preaxial Polydactyly & Triphalangeal Thumb

Unaffected



Affected



CTAACACT **TAAGGAAGTGATT** AAGCAGCA

TBX5/6
EN1
CDX2
PITX1/2
PBX1
SHOX2
HOXA9, B8,
C4/5/6/8/9

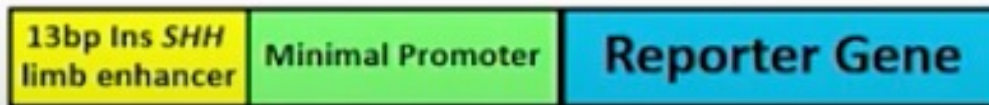


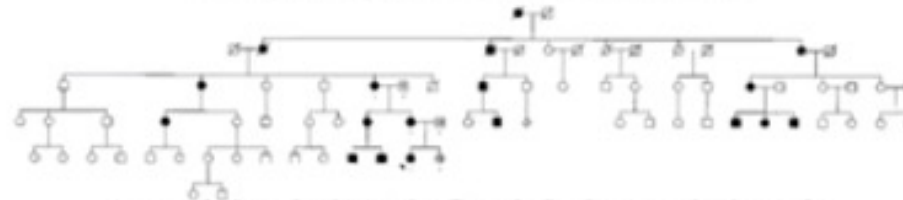
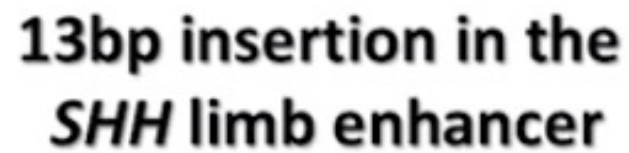
Laurell et al. *Human Mutation* 2012

Differential enhancer Assay

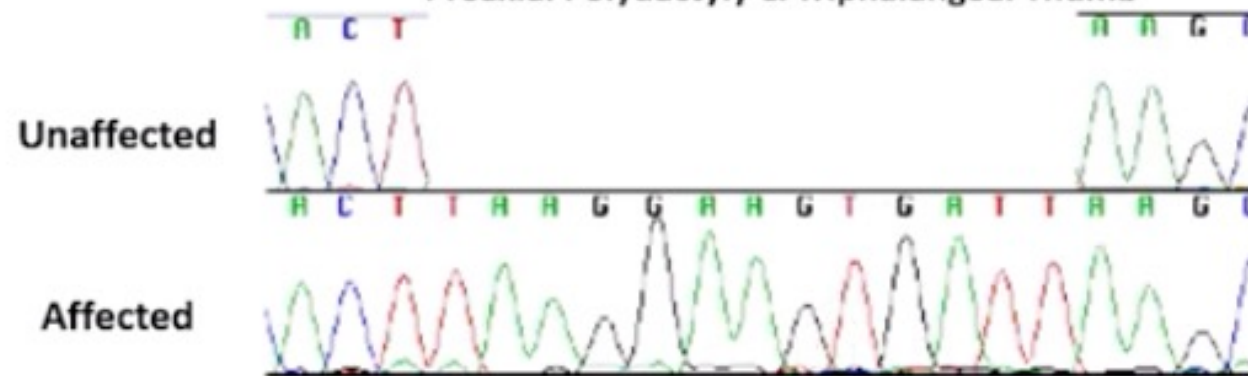


Difference?





Preaxial Polydactyly & Triphalangeal Thumb



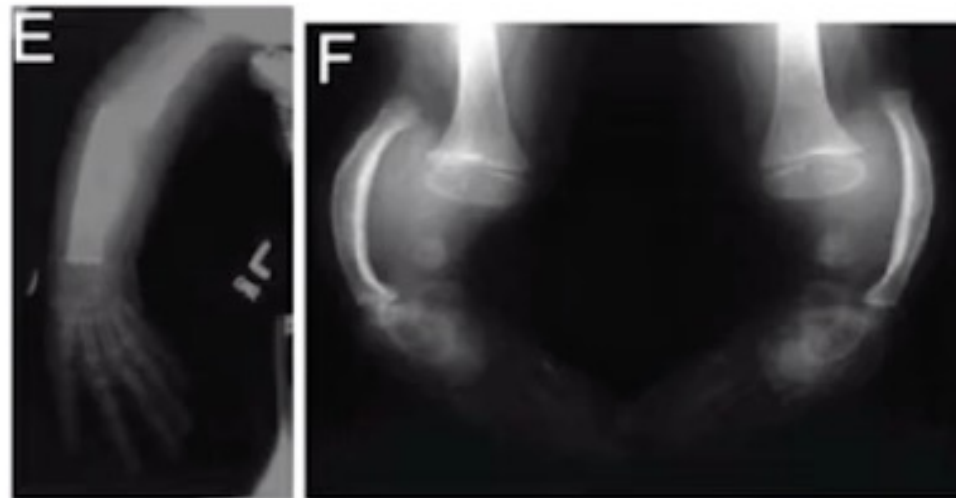
TBX5/6
EN1
CDX2
PITX1/2
PBX1
SHOX2
HOXA9, B8,
C4/5/6/8/9

CTAACACTTAAGGAAGTGATTAAAGCAGCA

Laurell et al. *Human Mutation* 2012

Mutations in positions 402 & 404 cause a much more severe phenotype that includes the radioulnar bones and tibia

CCTTGTACTGTATTT
 ↑ ↑



VanderMeer et al. *Human Mutation* 2014

Deciphering the regulatory code

AGGAAGGGAAAGCGCAAGAGAGAGC
GCACACGCACACACCCGCCGCGCGCA
CTCGCGCACGGACCCGCACGGGGAC
AGCTCGGAAGTCATCAGTTCCATGGGC
GAATGCTGCTGCTGGCGAGATGTCTGC
TGCTAGTCCTCGTCTCCTCGCTGCGGT
ATGCTCGGGACTGGCGTGCGGACCGG
GCAGGGGGTTTCGGGAAGAGGAGGCA
CCCCAAAAGCTGATACACAAGCCTAC
AAGCAGTTTATCCCCAATGTGGCCGAG
AAGACCCTAGGCGCCAGCGGAAGGTA
TGAAGGGAAGATCTCCAGAACTCCG
AGCGATTTAAGGAACTCACCCCCAATT
ACAACCCCGACATCATATTTAAGGATG
AAGAAAACACCGGAGCGGACAGGCT
GATGACTCAGTAGGAACCCAGCGCCG

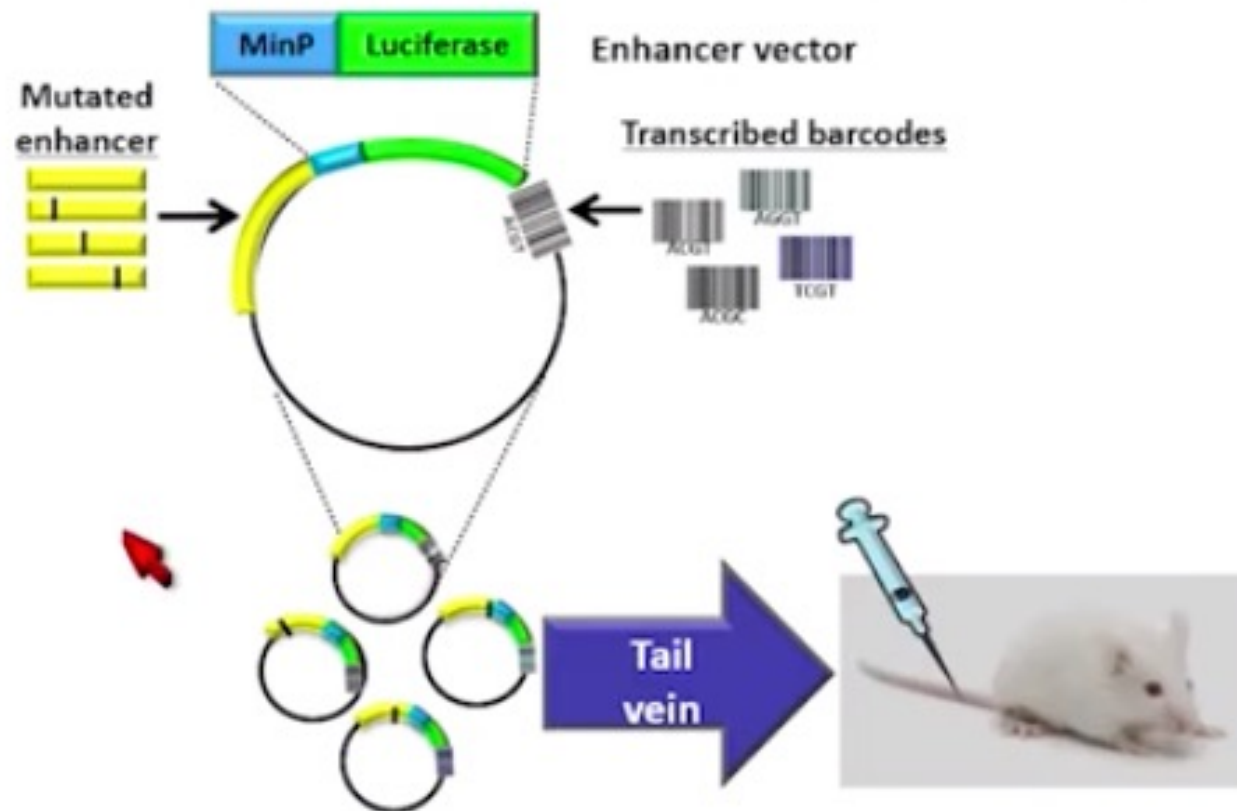
Min.
Pr.

GFP



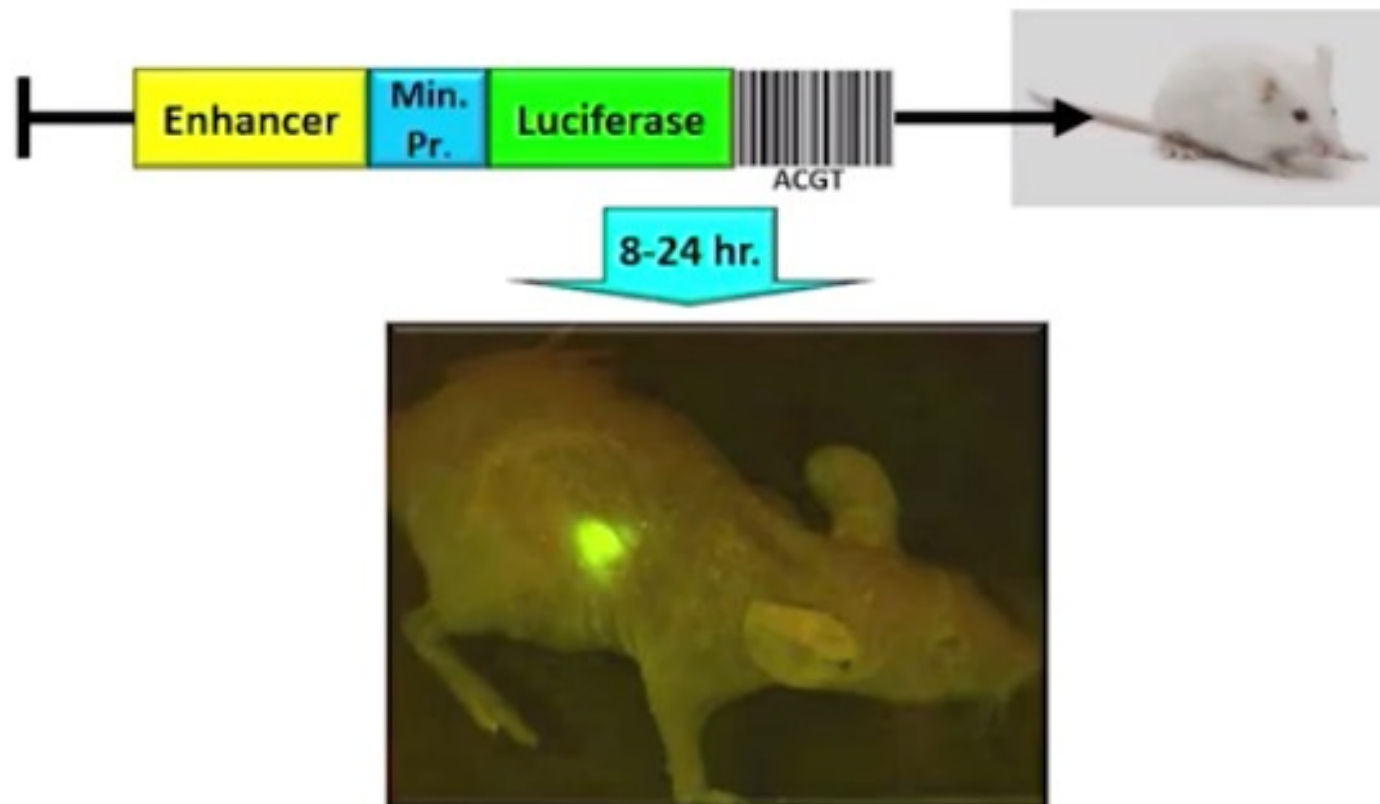
1. Deconstruction: mutate every base.
2. Construction: build a language from scratch.

Saturation mutagenesis massively parallel reporter assay (MPRA)



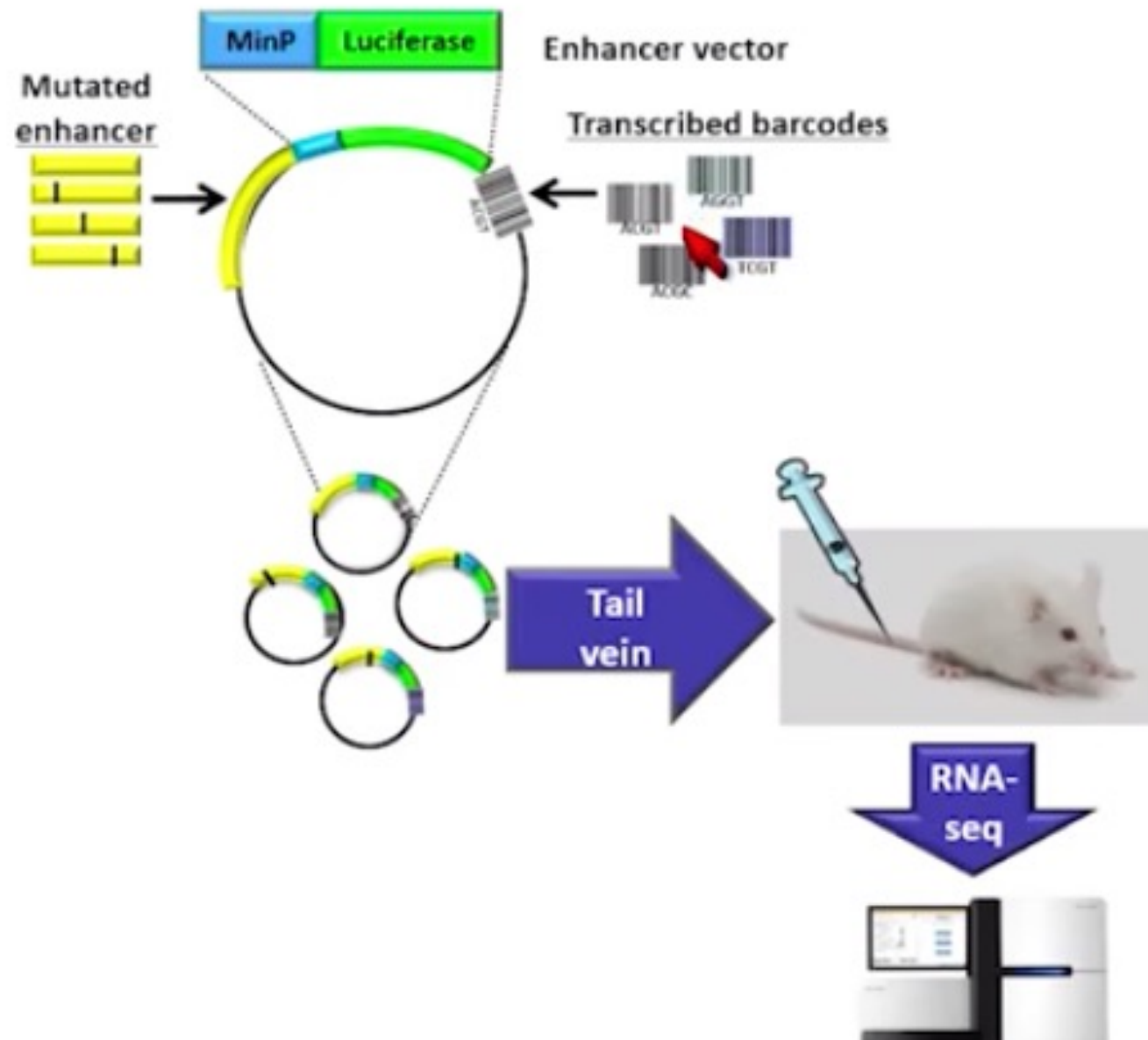
Hydrodynamic Tail Vein Injection

- Large volume: 10% of mouse weight (30gr→3ml).
- Short time: 4-8 seconds.

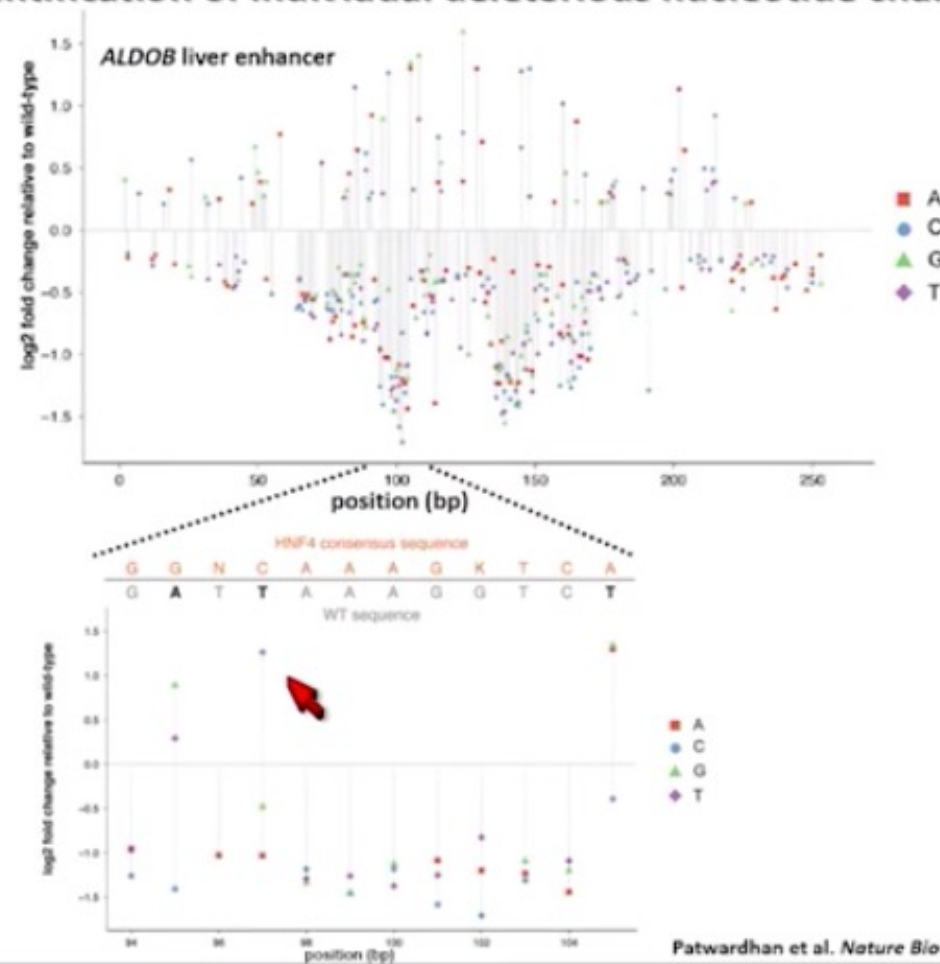


Yang M. et al. *PNAS* 2000, 97:12278-12282

Saturation mutagenesis massively parallel reporter assay (MPRA)



Identification of individual deleterious nucleotide changes



Deciphering the regulatory code

AGGAAGGGAAAGCGCAAGAGAGAGC
 GCACACGCACACACCCGCCGCGCGCA
 CTCGCGCACGGACCCGCACGGGGAC
 AGCTCGGAAGTCATCAGTTCCATGGGC
 GAATGCTGCTGCTGGCGAGATGTCTGC
 TGCTAGTCCTCGTCTCCTCGCTGCGGT
 ATGCTCGGGACTGGCGTGCGGACCGG
 GCAGGGGGTTCGGGAAGAGGAGGCA
 CCCCAAAAAGCTGATACACAAGCCTAC
 AAGCAGTTTATCCCCAATGTGGCCGAG
 AAGACCCTAGGCGCCAGCGGAAGGTA
 TGAAGGGAAGATCTCCAGAAACTCCG
 AGCGATTTAAGGAACTACCCCCAATT
 ACAACCCCGACATCATATTTAAGGATG
 AAGAAAACACCGGAGCGGACAGGCT
 GATGACTCAGTAGGAACCCAGCGCCG

Min.
Pr.

GFP



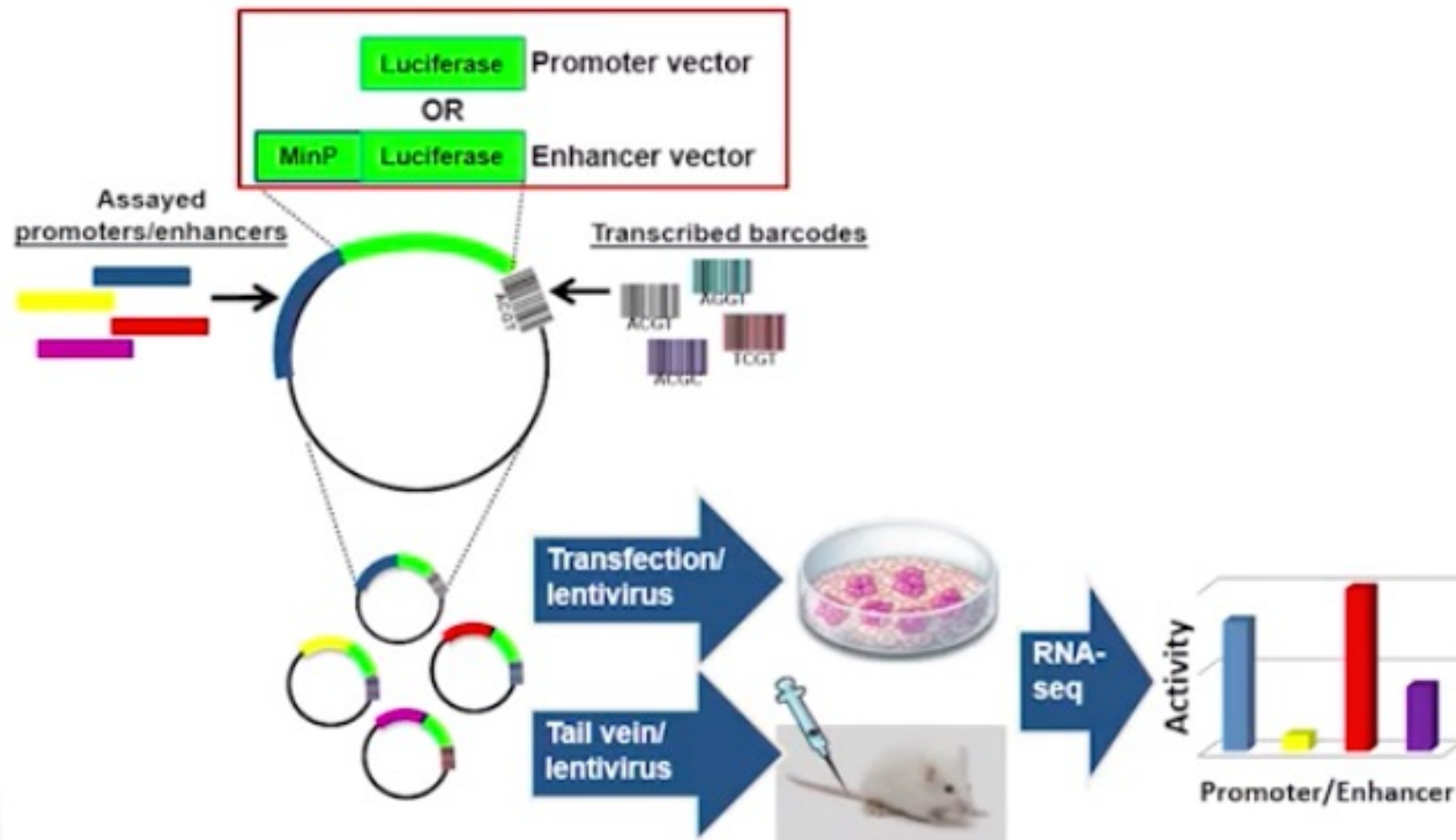
2
 1
 0
 5' G A T A C A 3'

Min.
Pr.

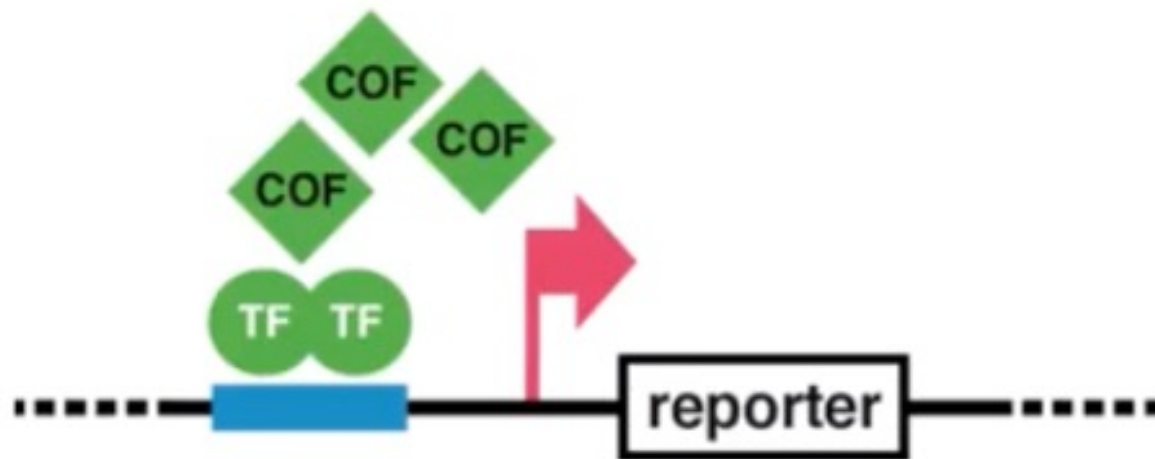
GFP



Massively Parallel Reporter Assays



Studying transcriptional regulation in defined reporter assays



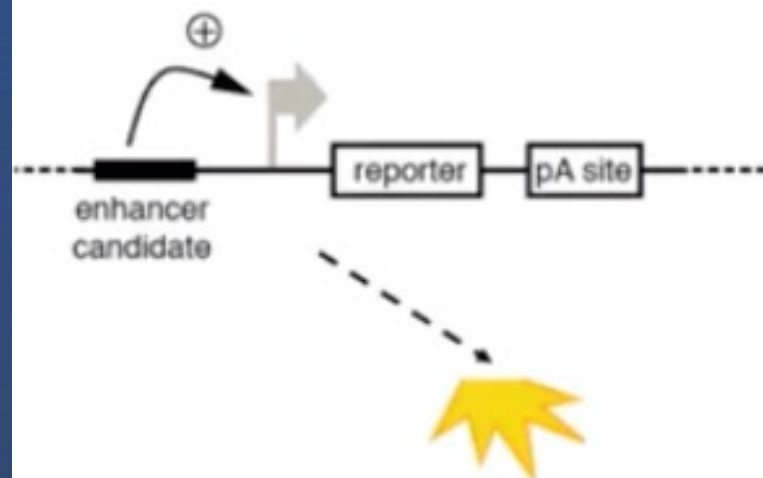
Identification and characterization of:

- **enhancers**
- **promoters**
- **TFs and cofactor proteins**

STARR-Seq

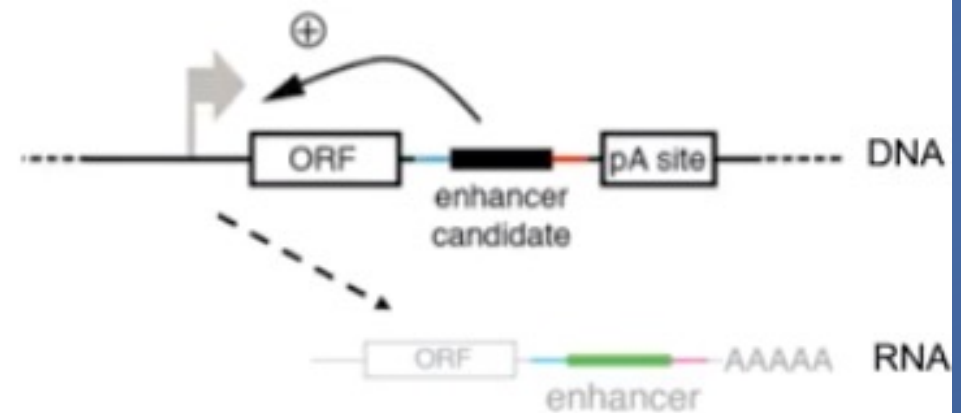
Active enhancers transcribe themselves

Luciferase reporter assay



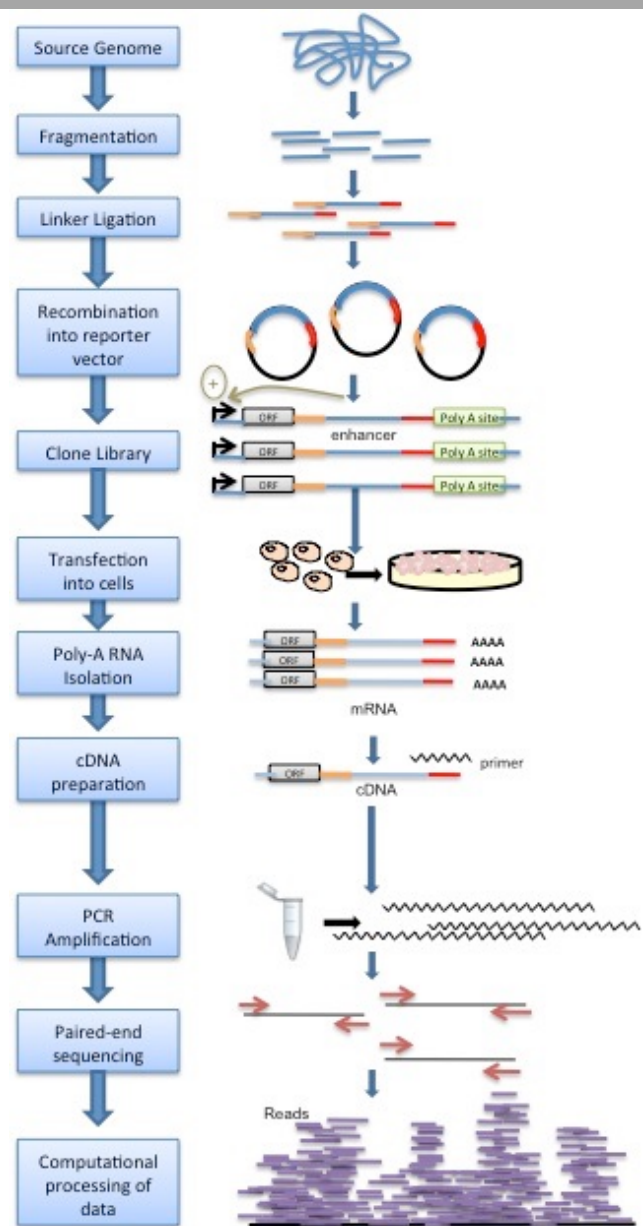
- + activity based
- + quantitative
- low throughput (one-by-one)

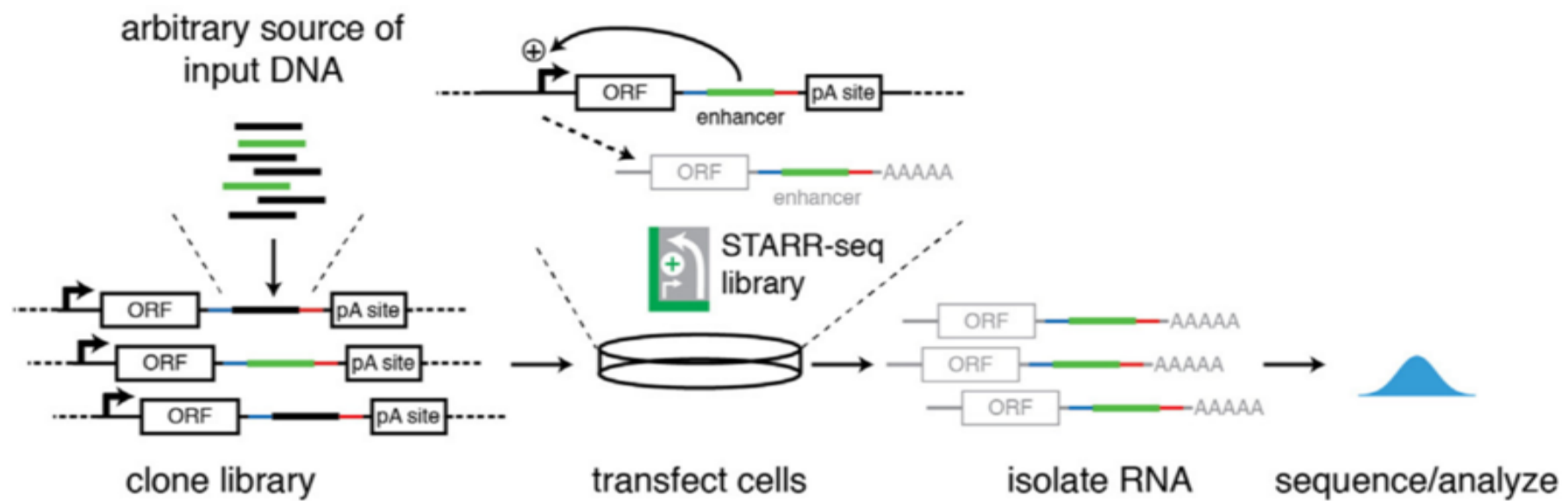
STARR-seq coupling activity to sequence

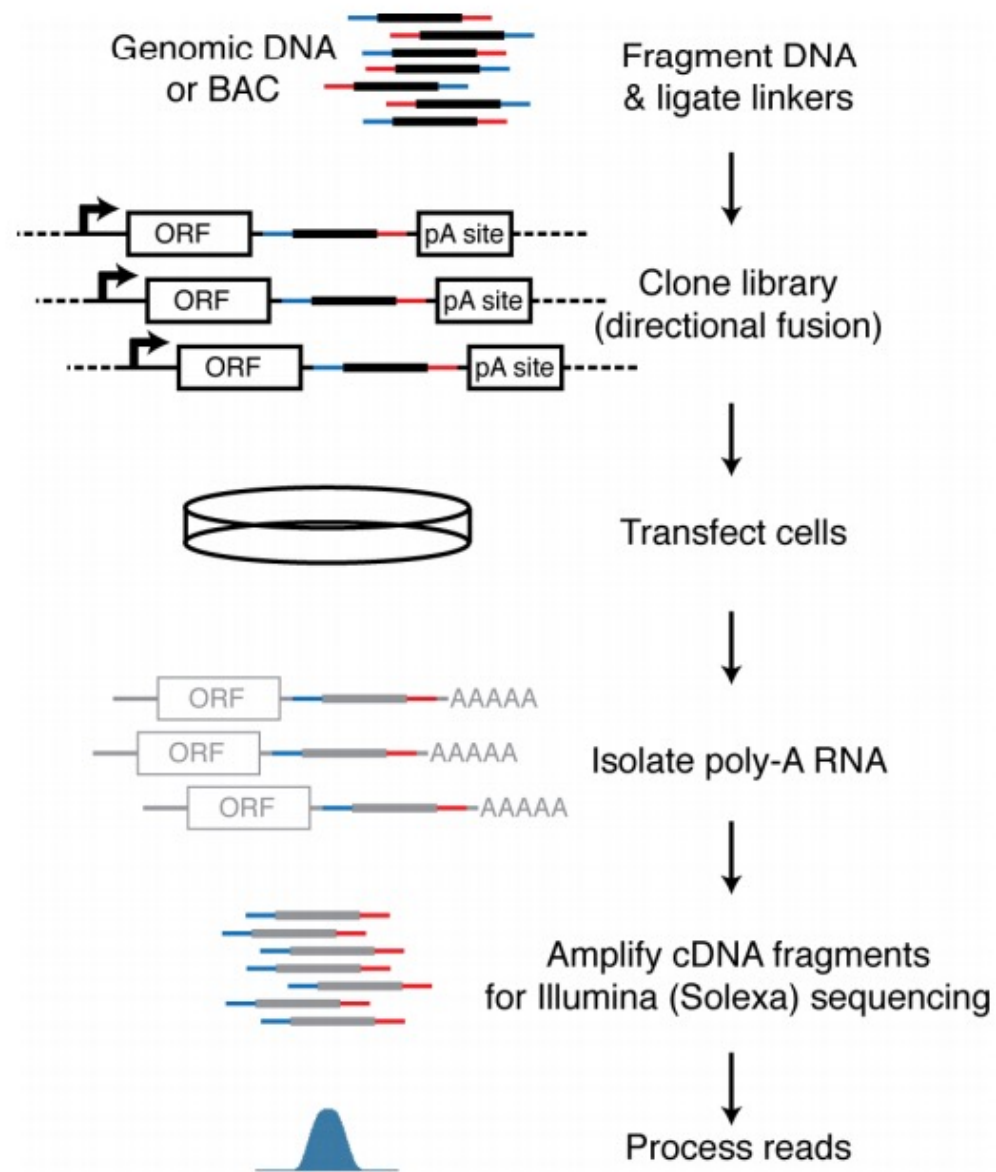


- + activity based
- + quantitative
- + genome-wide

Arnold *et al.*, Science 2013



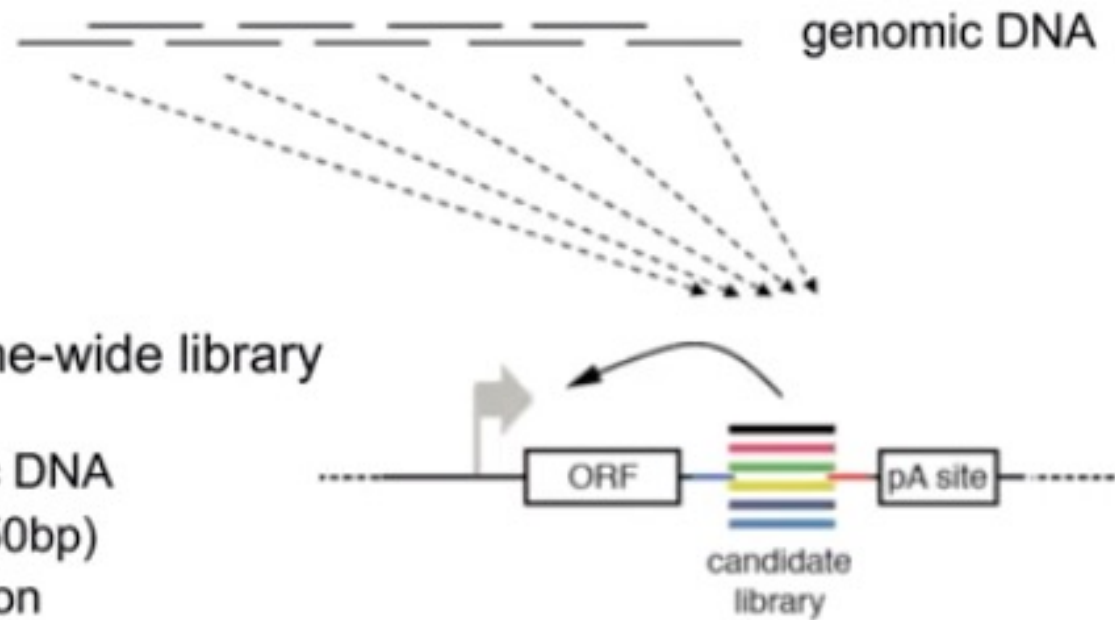




Cloning of highly complex genome-wide candidate libraries

Generation of a genome-wide library

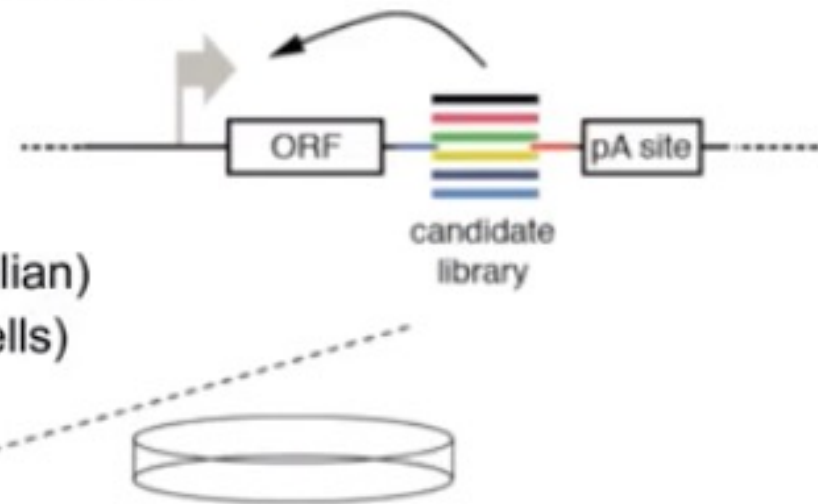
- sonication of genomic DNA
 - size selection (500-750bp)
 - Illumina adapter ligation
 - cloning into STARR-seq vector in batch
- 10^8 - 10^9 different fragments/clones



Large scale transfection using the MaxCyte STX electroporation system

Transfection of highly complex reporter libraries

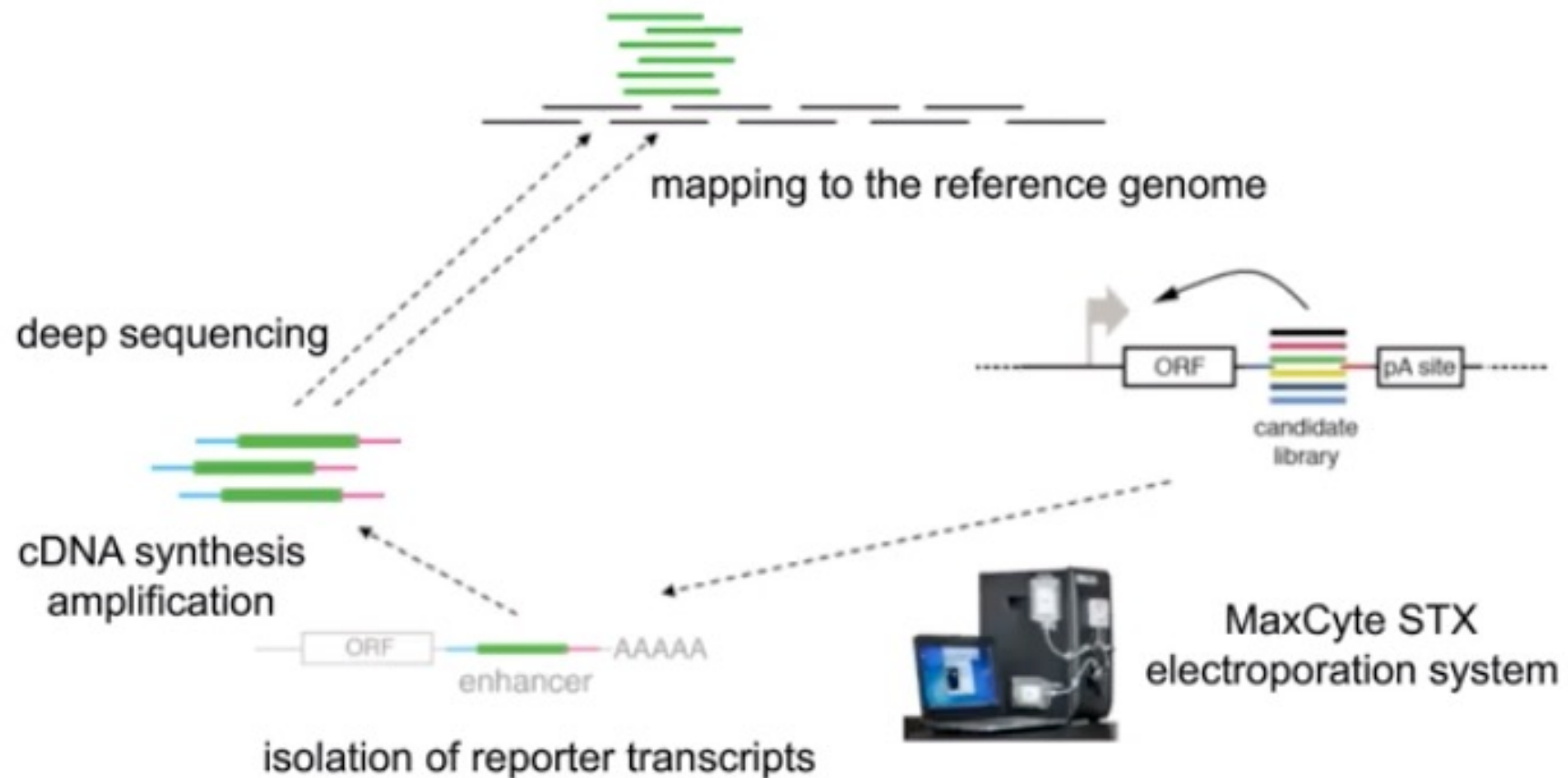
- $> 2-5 \times 10^9$ cells
- high & reliable transfection efficiency
- high & reliable cell viability
- wide range of cell lines (insect & mammalian)
- high amount of DNA ($20\mu\text{g DNA}/1 \times 10^7$ cells)



MaxCyte STX
electroporation system

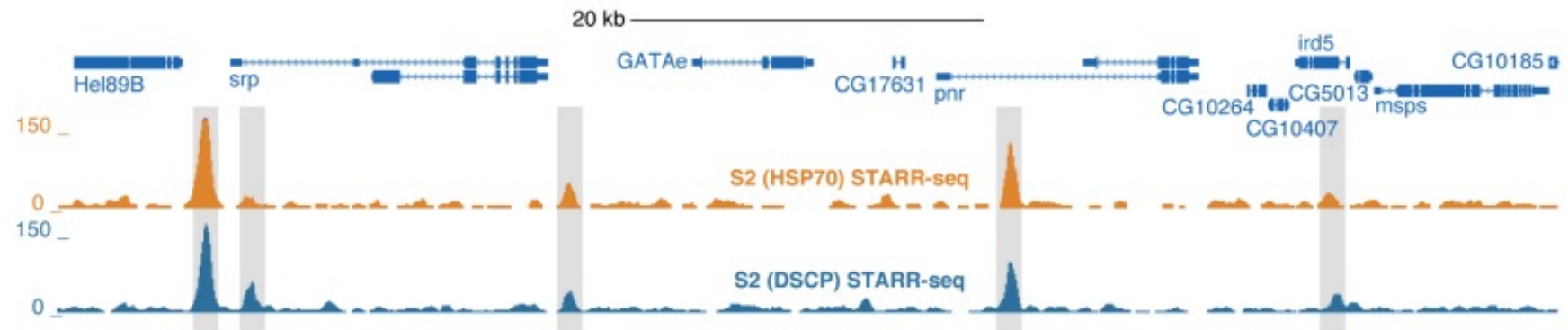
Arnold *et al.*, Science 2013

STARR-seq – measuring enhancer activity by reporter transcript abundance

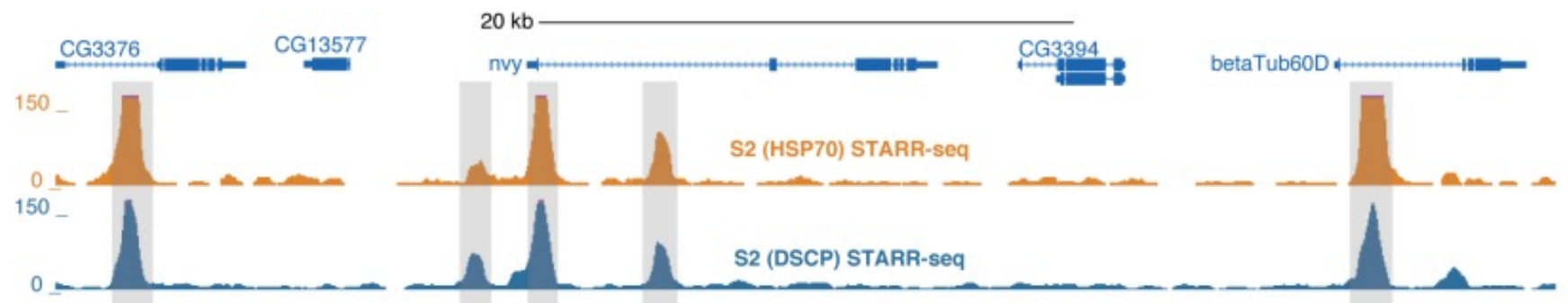


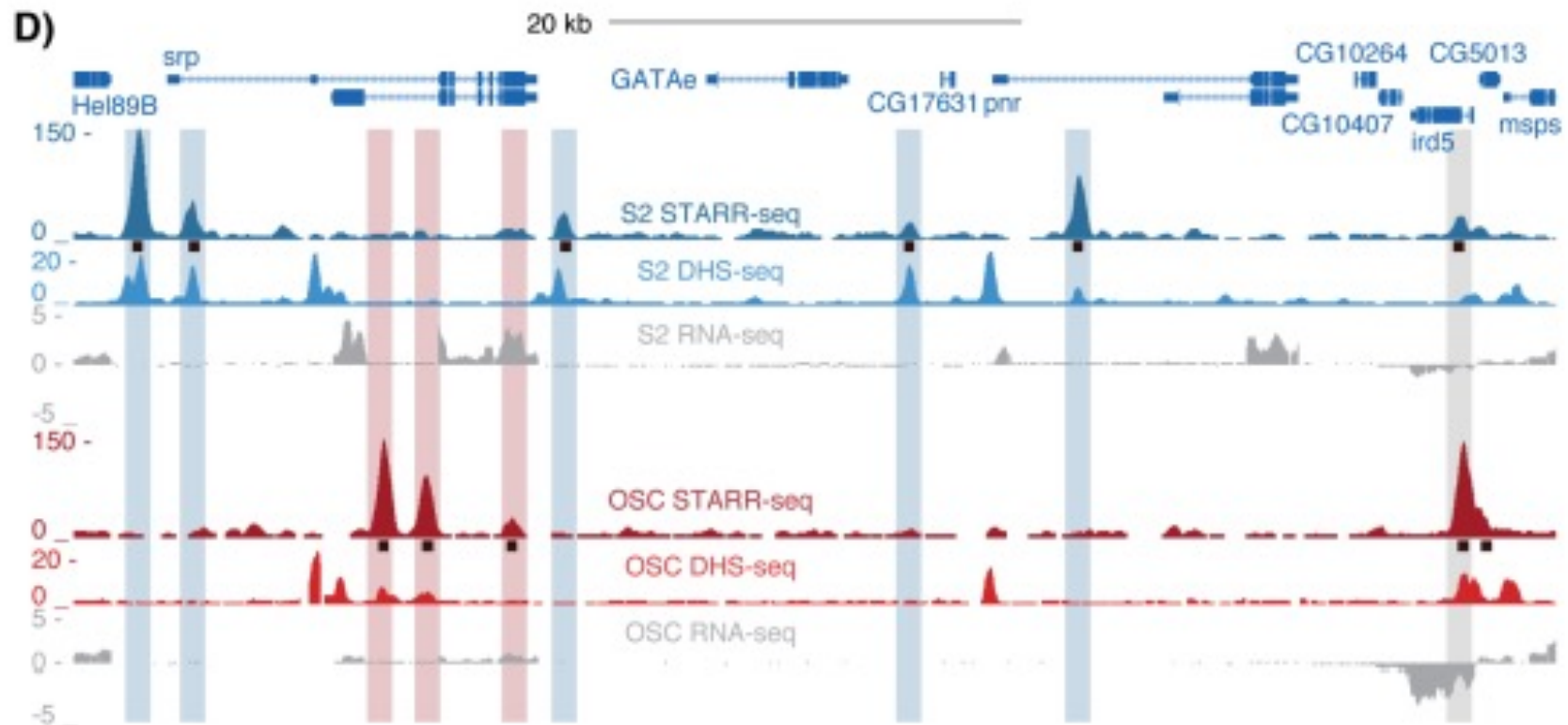
Arnold *et al.*, Science 2013

E)

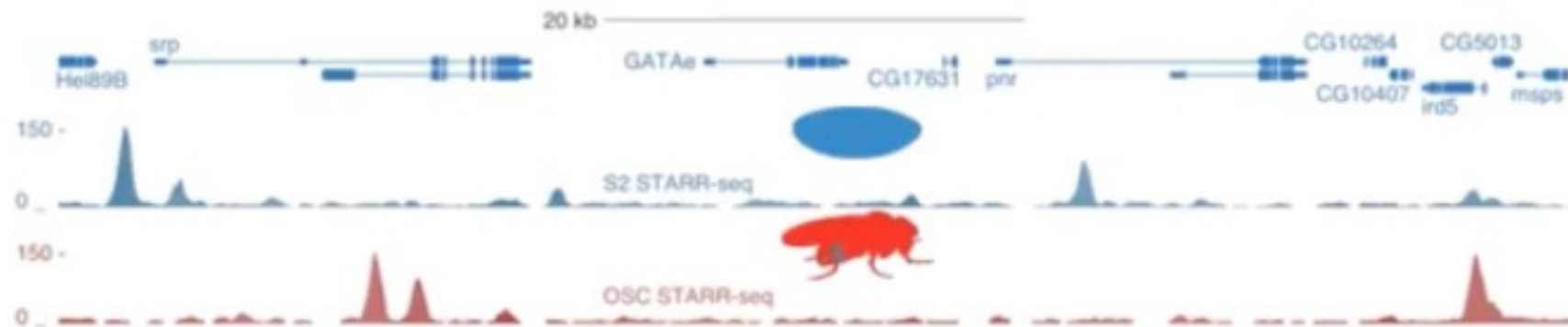


F)





STARR-seq – genome-wide quantitative enhancer activity maps

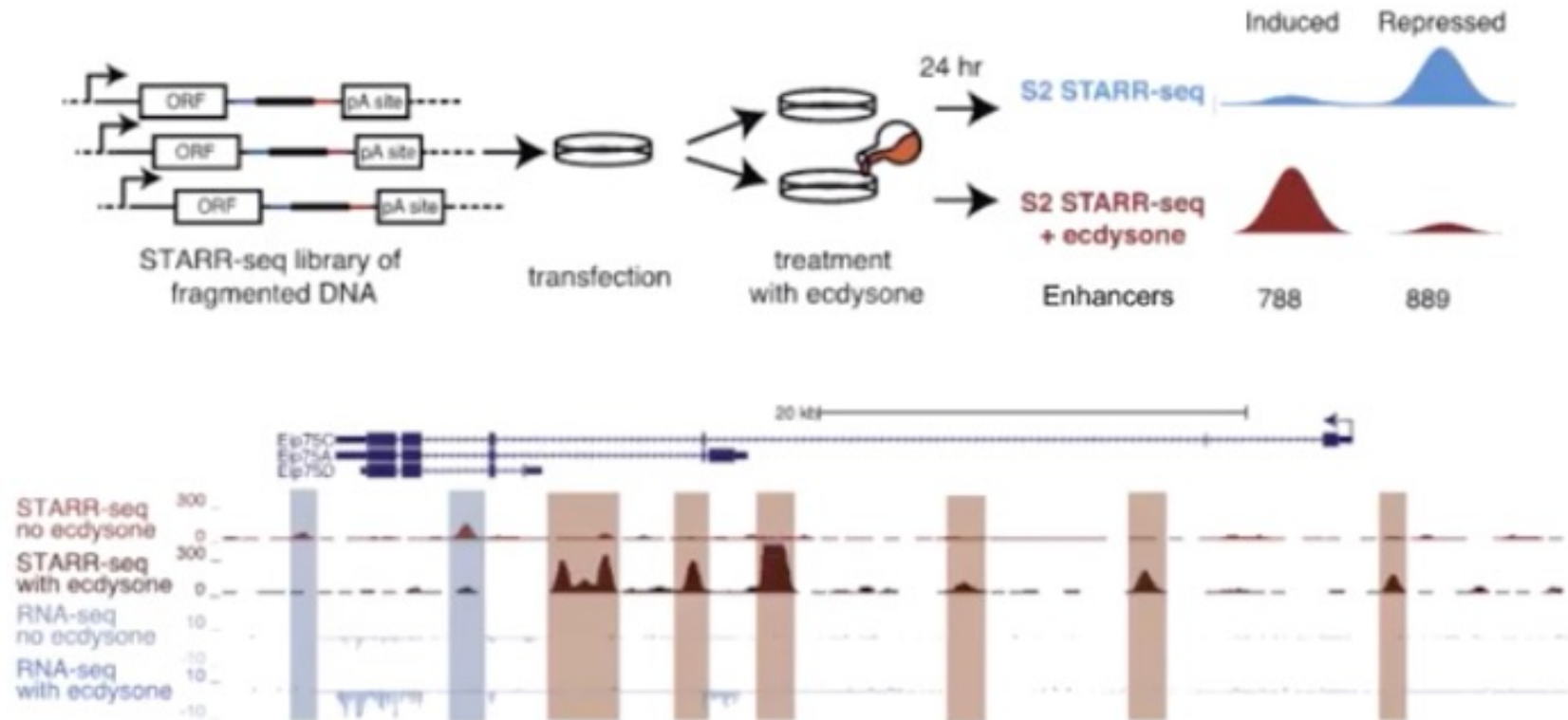


- cell-type specific
- applicable genome-wide
- quantitative (high agreement with luciferase)
- enhancers function *in vivo* (11/13)
- insect & mammalian cell lines



Arnold *et al.*, Science 2013

Hormone/Signaling-responsive enhancers in *Drosophila*

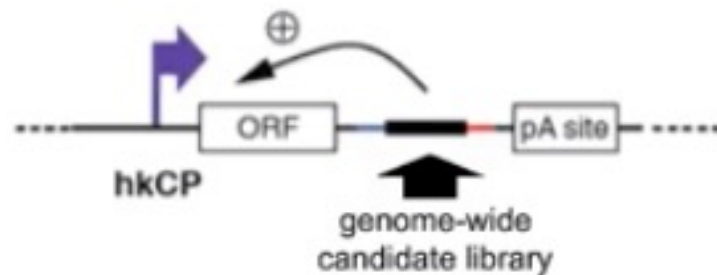


Shlyueva *et al.*, Molecular Cell 2014

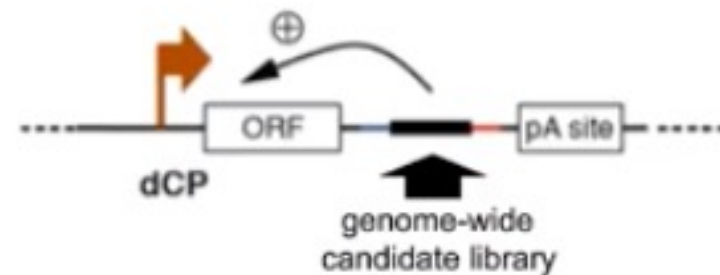
Identification of Wnt/Wingless-responsive enhancers in *Drosophila* Kc cells

Franz & Shlyueva *et al.*, PLOS Genetics 2017

STARR-seq – enhancer activity towards different core promoters



housekeeping core promoter
(ribosomal protein gene)



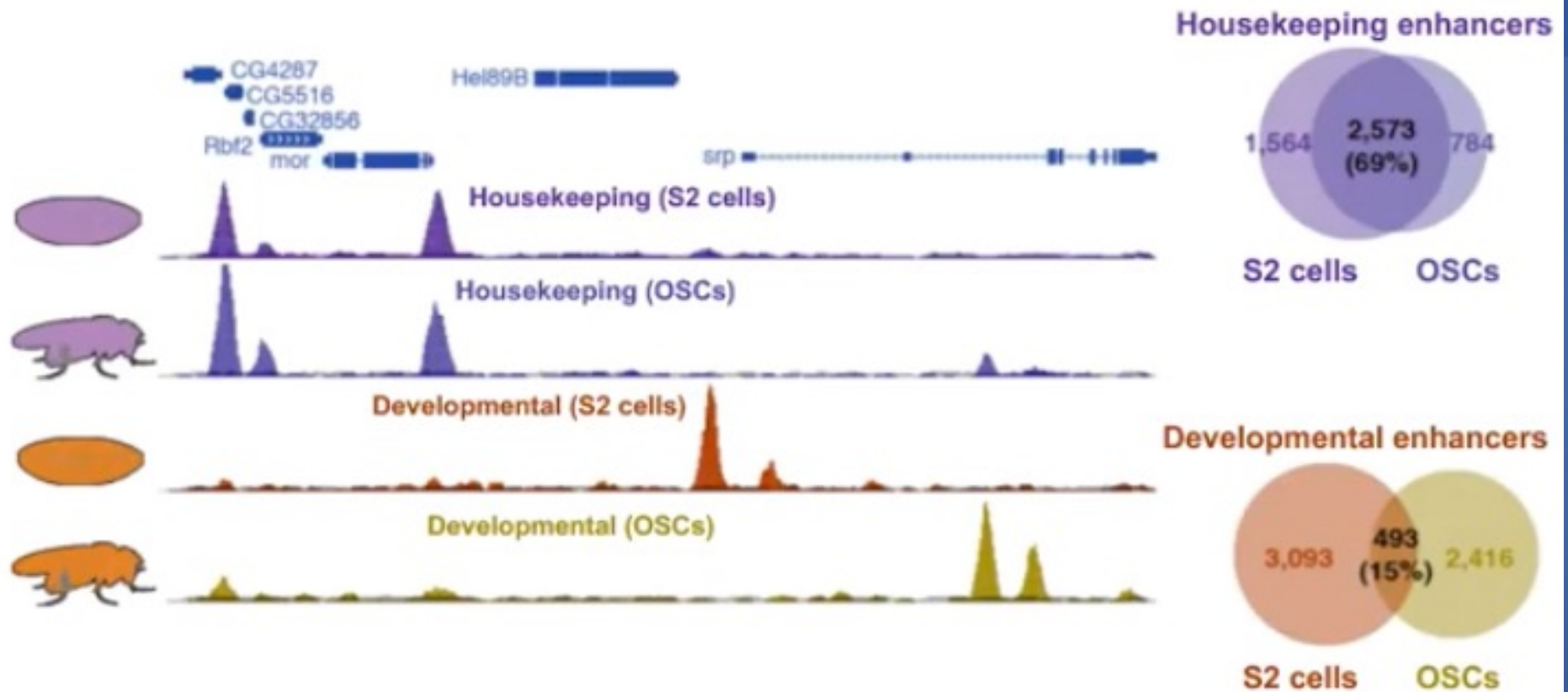
developmental core promoter
(~ TF *even-skipped*)

→ which genomic fragments activate transcription from which core promoter?
otherwise entirely defined & constant context

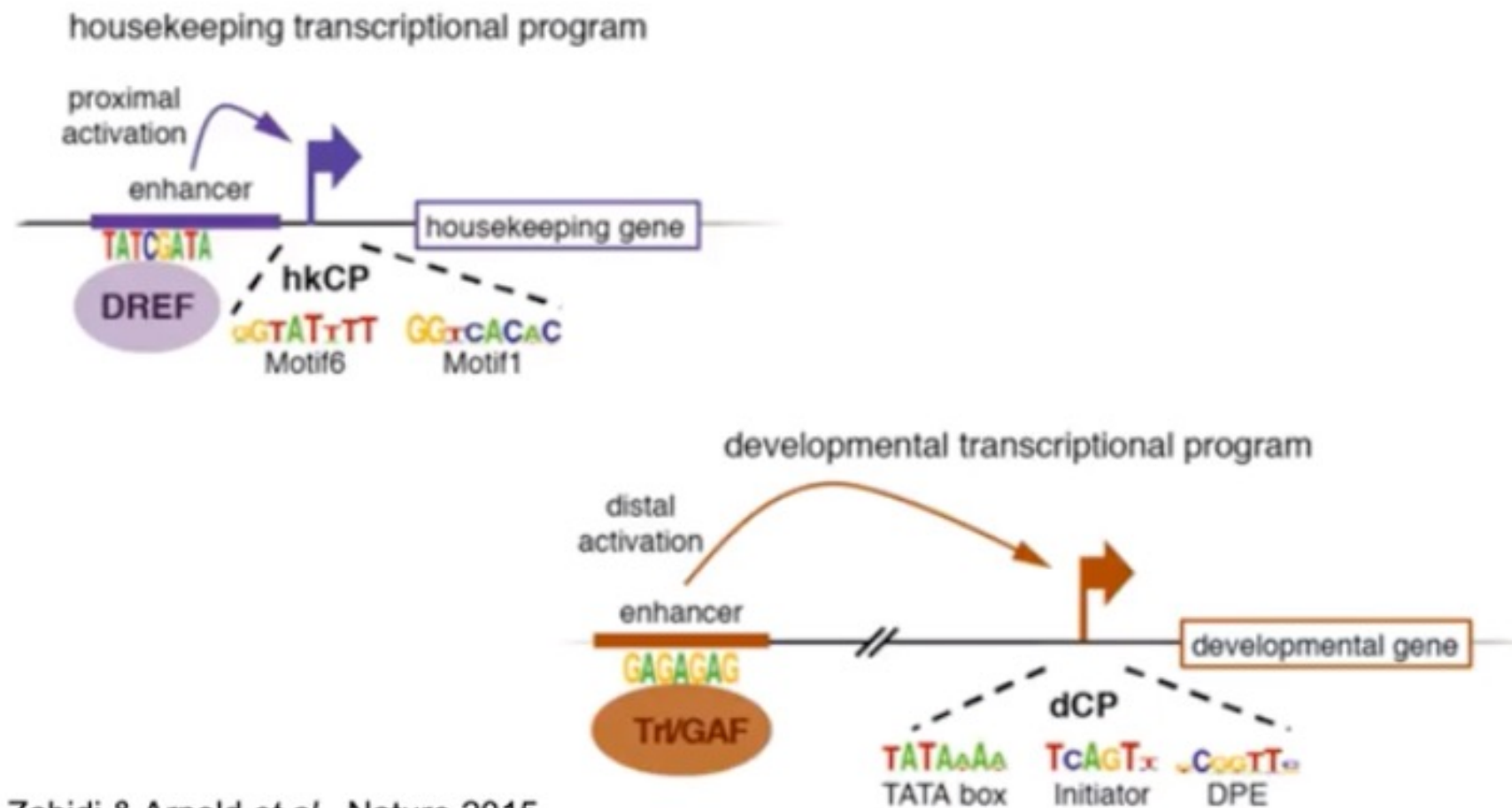
Developmental and housekeeping core promoters are activated by different enhancers in *Drosophila* S2 cells



Housekeeping enhancers are shared across cell types

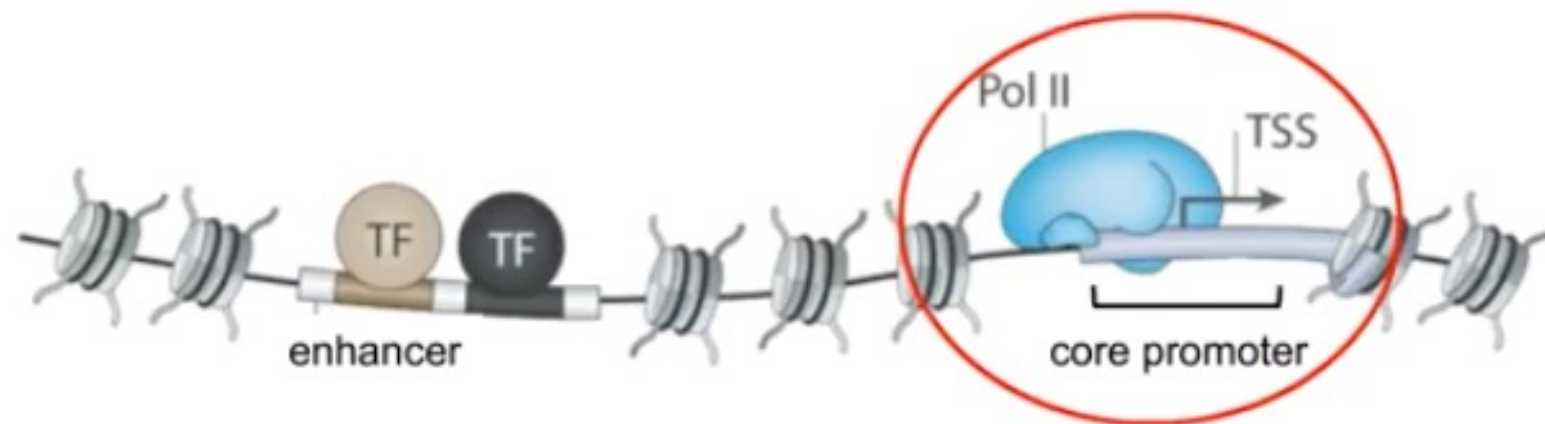


Enhancer – core promoter specificity decouples developmental from housekeeping gene regulation



Zabidi & Arnold *et al.*, Nature 2015

Influence of core promoters on transcription initiation & levels?



- can any sequence in the genome function as CP?
- what is the contribution of core promoters to gene expression levels?
- do core promoters respond differentially to certain types of enhancers?